

Full length article

Autoregressive inverse design of disordered metamaterials for target nonlinear response

Dazhi Zhao^a, Yujie Xiang^a, Peng Zhang^a, Ning Liu^a, Xianqiao Wang^b, Keke Tang^{a,c,*}^a School of Aerospace Engineering and Applied Mechanics, Tongji University, Shanghai 200092, China^b School of Environmental, Civil, Agricultural & Mechanical Engineering, University of Georgia, Athens, GA 30602, USA^c Key Laboratory of AI-aided Airworthiness of Civil Aircraft Structures, Civil Aviation Administration of China, Tongji University, Shanghai 200092, China

ARTICLE INFO

Keywords:

Data-driven design
Disordered metamaterials
Structural connectivity
Autoregressive model
Inverse design
Nonlinear response

ABSTRACT

The pursuit of novel mechanical metamaterials faces a core dilemma. While introducing architectural disorder unlocks unprecedented design space, it jeopardizes the structural connectivity essential for mechanical integrity. We introduce a data-driven framework to resolve this critical trade-off in programming nonlinear responses. Our approach pairs a generative autoregressive model for inverse design with a surrogate model for rapid forward prediction. The autoregressive model learns implicit connectivity rules from data to sequentially build valid structures targeting specific performances. We demonstrate that this method successfully designs disordered metamaterials with target nonlinear properties on demand. Analysis of the model's internal mechanism reveals its learned strategies for maintaining connectivity, highlighting a critical yet controllable trade-off between performance and connectivity. The framework exhibits strong interpolation and extrapolation capabilities, generating novel designs that outperform those in the training set. By providing a powerful and generalizable design tool, this work establishes a pathway to reliably meet complex functional requirements with disordered metamaterials, moving them from conceptual appeal toward practical viability.

1. Introduction

Mechanical metamaterials have redefined the boundaries of materials science by enabling unprecedented performance derived from their structural architecture rather than the constituent materials [1–4]. Consequently, inverse design strategies have become indispensable tools to identify specific global architectures that yield tailored mechanical performance [5–10]. While foundational research primarily focuses on periodic architectures [11–15], recent TPMS-based hybrid designs [16] and bi-material lattice strategies [17] demonstrate how geometric tailoring and material heterogeneity within periodic frameworks enhance thermo-mechanical performance. Nevertheless, relaxing strict periodic constraints further enlarges the design space and enables a broader spectrum of achievable nonlinear mechanical responses [18, 19]. This is particularly evident when complex nonlinear responses such as buckling [20] and self-contact [21] are required for advanced functionalities such as soft robotics [22–24] and impact energy absorption [25]. To expand the design space and enable the programming of these specific deformation paths, architectures have evolved toward disordered configurations realized by assembling mixed unit cells [26–28].

To physically realize these disordered configurations, algorithmic strategies such as the Virtual Growth Algorithm (VGA) have emerged to establish the geometric foundation [29,30]. By stochastically assembling unit cells subject to strict local compatibility rules, these methods inherently guarantee connectivity and generate valid geometric domains [29]. However, acting primarily as modeling tools, identifying a specific architecture that matches a target mechanical performance remains a formidable challenge. The immense discrete design space, coupled with the high cost of iterative nonlinear finite element simulations, renders combinatorial search and traditional optimization computationally prohibitive for these complex systems [31,32].

Data-driven approaches have subsequently emerged as a necessary alternative to overcome these computational bottlenecks [33–46]. Recent advancements have demonstrated the efficacy of these methods in optimizing B-spline honeycombs for energy absorption, designing lattice metamaterials with customized stress-strain responses, and orchestrating the forward prediction and inverse design of 4D printed hierarchical architectures with arbitrary shapes [47–52]. This creates a hierarchical requirement for an efficient design framework. First, a high-speed surrogate model is needed to replace expensive simulations

* Corresponding author.

E-mail address: kktang@tongji.edu.cn (K. Tang).<https://doi.org/10.1016/j.tws.2026.114793>

Received 15 January 2026; Received in revised form 26 February 2026; Accepted 8 March 2026

Available online 11 March 2026

0263-8231/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

for forward prediction, providing the massive dataset required to train the inverse model [53–55]. Second, and more critically, this generative model is required to execute the inverse mapping, directly predicting structural configurations from desired performance targets [56,57]. This enhanced mapping capability effectively unlocks the exploration of disordered architectures allowing for the rapid evaluation required by inverse design frameworks.

Yet, a fundamental connectivity problem arises, when integrating deep learning with disordered assembly. While algorithms like the VGA inherently ensure connectivity during data generation [29], standard generative models typically treat structures as pixels or probability vectors. For instance, Xiang et al. employed deep learning for multi-material unit cells but did not effectively resolve the topological mismatches arising from mixing diverse geometric units [58]. Similarly, Zhou et al. introduced a probabilistic method for growth-inspired materials, yet their implementation lacked the sequential conditioning required to strictly enforce local compatibility [59]. This disregard for local positional constraints results in frequent geometric violations, indicating that the connectivity problem remains a critical hurdle for generative methods.

Current research addressing this challenge reveals a clear divide between iterative optimization strategies and end-to-end generative methods. One stream utilizes iterative strategies based on rule-based algorithms or tiling optimizations to ensure connectivity. For instance, Wang et al. assembled disordered metamaterials via database-driven tiling [31,60], while Jia et al. integrated the VGA into an inverse optimization framework [61,62]. However, these combinatorial methods often struggle to efficiently navigate the high-dimensional landscape associated with complex nonlinear targets, restricting their ability to find optimal solutions beyond local minima [63]. The other stream leverages advanced deep learning to capture sophisticated nonlinear responses. For example, Bastek et al. employed video diffusion models to successfully predict complex behaviors like buckling and self-contact [64]. Yet, such data-driven approaches either remain confined to periodic architectures or fail to resolve the connectivity problem in disordered assemblies [65–67]. This highlights a distinct division: iterative strategies guarantee connectivity but lack efficiency for nonlinear design, while current generative models offer efficiency but struggle to ensure the connectivity of disordered configurations.

Addressing this gap requires reconciling the global target for nonlinear response with the local imperative of geometric connectivity. The emergence of autoregressive paradigms offers a promising strategy for mechanical design by framing construction as a sequential decision-making process, where each choice is conditioned on both performance targets and prior actions [68–73]. For instance, Buehler et al. applied Large Language Models (LLMs) to text-based design [74–76], while Zhang et al. utilized this approach to ensure connectivity in porous materials [77]. Similarly, Maurizi et al. used Reinforcement Learning (RL) to progressively construct trusses with target nonlinear responses [78]. However, while graph-based connectivity depends on simple node-edge relationships, mixed unit cell assembly involves complex multi-directional compatibility constraints that are harder to enforce. Consequently, applying this paradigm to the strict connectivity

constraints of disordered metamaterials composed of mixed unit cells has yet to be fully established.

To clearly position our work within the existing landscape, Table 1 summarizes the key differences between the proposed autoregressive framework and other prevailing inverse design paradigms.

To overcome the limitations outlined above, we present a data-driven inverse design framework that addresses the established hierarchical requirements, as shown in Fig. 1. We model inverse design as a conditional sequence generation task. To resolve the connectivity problem within the inverse mapping, we employ a generative Autoregressive (AR) model that takes the target nonlinear response as a prompt to progressively predict the next unit cell, thereby inherently satisfying the connectivity. Simultaneously, to eliminate the computational bottleneck of forward prediction, a Fourier Neural Operator (FNO) is integrated as a high-speed surrogate model. This approach allows for efficient mapping of the design space and establishes a robust pathway for the on-demand design of disordered nonlinear metamaterials.

The remainder of the paper is organized as follows. Section 2 details our methodology, from data generation to the model architectures for forward prediction and inverse design. Section 3 presents the results, analyzing the inverse model's decision process, performance trade-offs, and overall interpolation and extrapolation capabilities. Section 4 concludes the paper with a summary and discussion of future work.

2. Methodology

2.1. Metamaterial structure generation via virtual growth algorithm

To build a diverse dataset for nonlinear inverse design, we employ the VGA [29]. This method is particularly suited for this task as it directly addresses the connectivity challenge by constructing structures based on a set of predefined connectivity rules, guaranteeing a collection of disordered but internally coherent metamaterials.

A candidate unit cell library is established, as shown in Fig. 2a. This library not only contains typical configurations with unique mechanical properties such as Chiral, Star, Re-entrant, Bowtie, and Diamond units, but also further broadens the design space by introducing their rotational variants. The unit cell library, assembled from these building blocks, covers a wide range of Poisson's ratios and provides the necessary structural diversity foundation for achieving complex and varied nonlinear responses. To allow these discrete geometric configurations to be processed by deep learning models, one-hot encoding is applied to every unique unit cell.

The selection of the unit cell library rests on four criteria: manufacturability, topological compatibility, mechanical substitutability, and the expansion of the design space. First, to ensure physical feasibility and mechanical robustness, we intentionally exclude geometries with excessively thin struts or acute angles. This ensures that any connectivity-valid assembly satisfies fabrication requirements. Second, the library prioritizes connectivity by covering diverse interface types, including node-edge, short-edge, and long-edge connections, which supports the assembly of disordered configurations. Third, the library provides mechanical redundancy by including distinct geometries that

Table 1
Comparison of the proposed framework with existing inverse design paradigms.

Design paradigm	Representative methods	Generation mode	Connectivity guarantee	Design efficiency
Periodic metamaterials				
Deep learning	GAN, Diffusion	One-shot (Pixel-based)	Low (Pixel/Probabilistic)	High (Direct inference)
Reinforcement learning (RL)	Graph-based RL	Sequential (Action-based)	Medium (Reward penalty)	Medium (Training intensive)
Disordered metamaterials				
Optimization	VGA-based optimization, topology optimization	Iterative (Trial-and-error)	High (Hard-coded rules)	Low (Repeated simulations)
Deep learning (Regression)	cVAE	One-shot (Pixel-based)	Low (Independent pixel)	High (Direct inference)
Deep learning (Classification)	MLP	One-shot (Vector-based)	Low (Independent vector)	High (Direct inference)
Proposed AR model	Transformer-based AR (Ours)	Sequential (Token-based)	High (Implicitly learned rules)	High (Autoregressive inference)

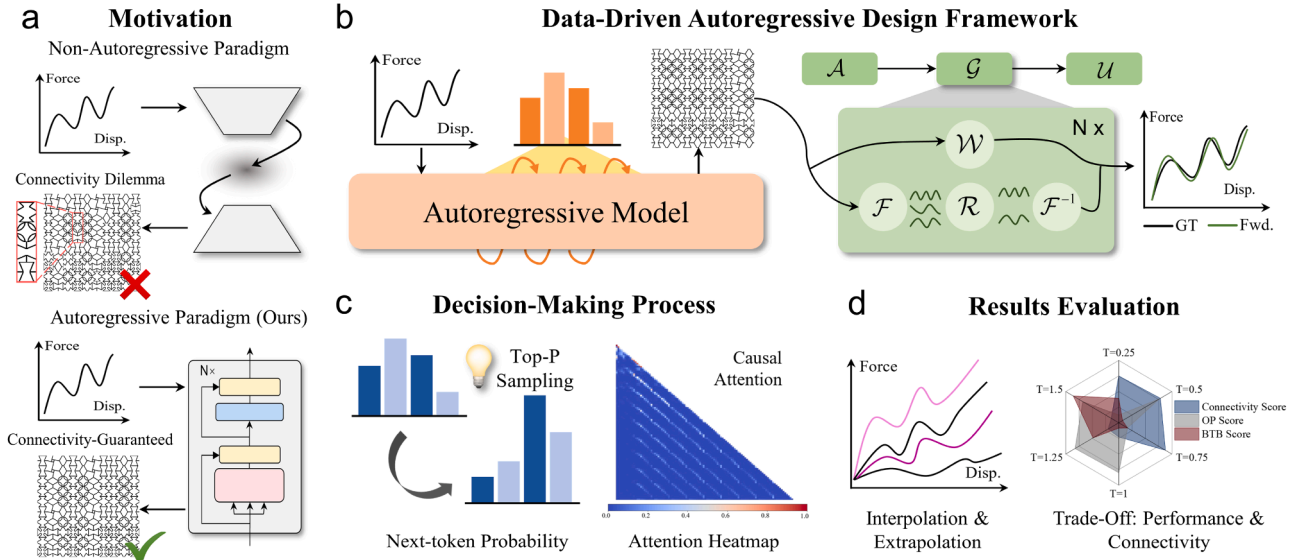


Fig. 1. Overview of the data-driven framework for the autoregressive inverse design of disordered metamaterials. (a) Motivation: Addressing the connectivity dilemma where architectural disorder often jeopardizes structural integrity in generative design. (b) Proposed Framework: To resolve this, a framework centered on an autoregressive (AR) model is proposed for sequential inverse mapping, integrated with a Fourier Neural Operator (FNO) surrogate for rapid performance prediction. (c) Decision-Making Process: To move beyond a black-box approach, the model's internal reasoning is analyzed. (d) Results Evaluation: Finally, the framework is validated through a comprehensive evaluation of its interpolation/extrapolation capabilities and the trade-off between target performance and structural connectivity.

yield similar stiffness properties. This characteristic allows the inverse model to identify functionally equivalent solutions, verifying that it learns underlying mechanical rules rather than rigid geometric mappings. Finally, the inclusion of units with varying deformation mechanisms, such as buckling and self-contact, ensures a broad range of achievable nonlinear responses.

The application of VGA depends on a set of strict connectivity rules which ensure the geometric compatibility between unit cells. Three types of valid connections are defined: node-edge connectivity, short-edge connectivity, and long-edge connectivity, strictly prohibiting the appearance of any irrational connectivity.

The specific growth process is displayed in Fig. 2b and proceeds iteratively on a 10 by 10 grid. The process begins with a random initial unit cell selection. Afterward, the algorithm will select the position with the minimum node entropy in the current grid to fill at each step. Minimum node entropy means that the position has the fewest valid choices and the strongest constraints. By selecting the solution of the current constrained node in order, the algorithm can generate a fully connected structure until the entire grid is filled. VGA also includes a backtracking mechanism. If a node's placement probability converges to zero, the algorithm executes iterative step retractions to reselect options.

2.2. Finite element analysis, mesh sensitivity, and quasi-static validation

It should be pointed out that the proposed framework is evaluated using finite element analysis as the numerical ground truth. These simulation models are based on verified hyperelastic constitutive laws and mesh convergence studies. This strategy validates the capability of the model to meet complex functional requirements before physical implementation.

The FE model setup is shown in Fig. 3a. An automated script is used to generate two-dimensional solids models from the VGA microstructures, resulting in a model size of 1000 mm by 1000 mm with plane strain condition applied. The top and bottom boundaries are kinematically coupled to two reference points. The lower reference point is fixed to prevent all displacement and rotation, while the upper reference point is subjected to a prescribed downward displacement of 450 mm. This displacement is applied smoothly using an amplitude curve. The

structures are meshed mainly with CPE4R. A global mesh size of 2.0 mm is used. To handle the large deformations, a general contact is defined to account for self-contact within the structure. The tangential behavior is modeled using a penalty formulation with a friction coefficient of 0.8. From the top reference point, the y-direction reaction force history is obtained to generate the nonlinear force-displacement curve characterizing the compressive response. To capture the geometric nonlinearity and represent the behavior of common rubber-like materials, we adopt an incompressible Mooney-Rivlin hyperelastic constitutive model. The corresponding strain energy density function is defined as:

$$W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) \quad (1)$$

where W is the strain energy density per unit reference volume. $C_{10} = 0.84 \text{ MPa}$ and $C_{01} = 0.21 \text{ MPa}$ are material constants, falling within the typical range for rubber [79]. $\bar{I}_1$ and $\bar{I}_2$ are the first and the second invariant of the deviatoric component of the left Cauchy-Green deformation tensor. Accordingly, the volumetric term in the strain energy function is neglected, consistent with the assumption of perfect incompressibility.

To ensure the reliability of the results, a mesh sensitivity analysis is conducted. As shown in Fig. 3b, the force-displacement curves exhibit convergence upon mesh refinement, confirming that the 2.0 mm mesh provides sufficient accuracy. The 2.0 mm mesh size is therefore selected to balance computational accuracy and efficiency. The quasi-static nature of the simulation was verified by comparing the internal and kinetic energies. As shown in Fig. 3c, the internal energy remains dominantly higher than the kinetic energy throughout loading, confirming that inertial effects are negligible.

For the 10×10 grid configuration, it is selected to serve as a balance between computational efficiency with physical fidelity. This scale ensures that the bulk mechanical behavior dominates over boundary-induced variations, thereby mitigating the influence of size effects on the global response. Furthermore, this resolution provides sufficient degrees of freedom to generate the diverse nonlinear buckling and contact behaviors required for the dataset.

The simulation setup employs finite boundary conditions with fixed and displacement constraints rather than periodic boundary conditions.

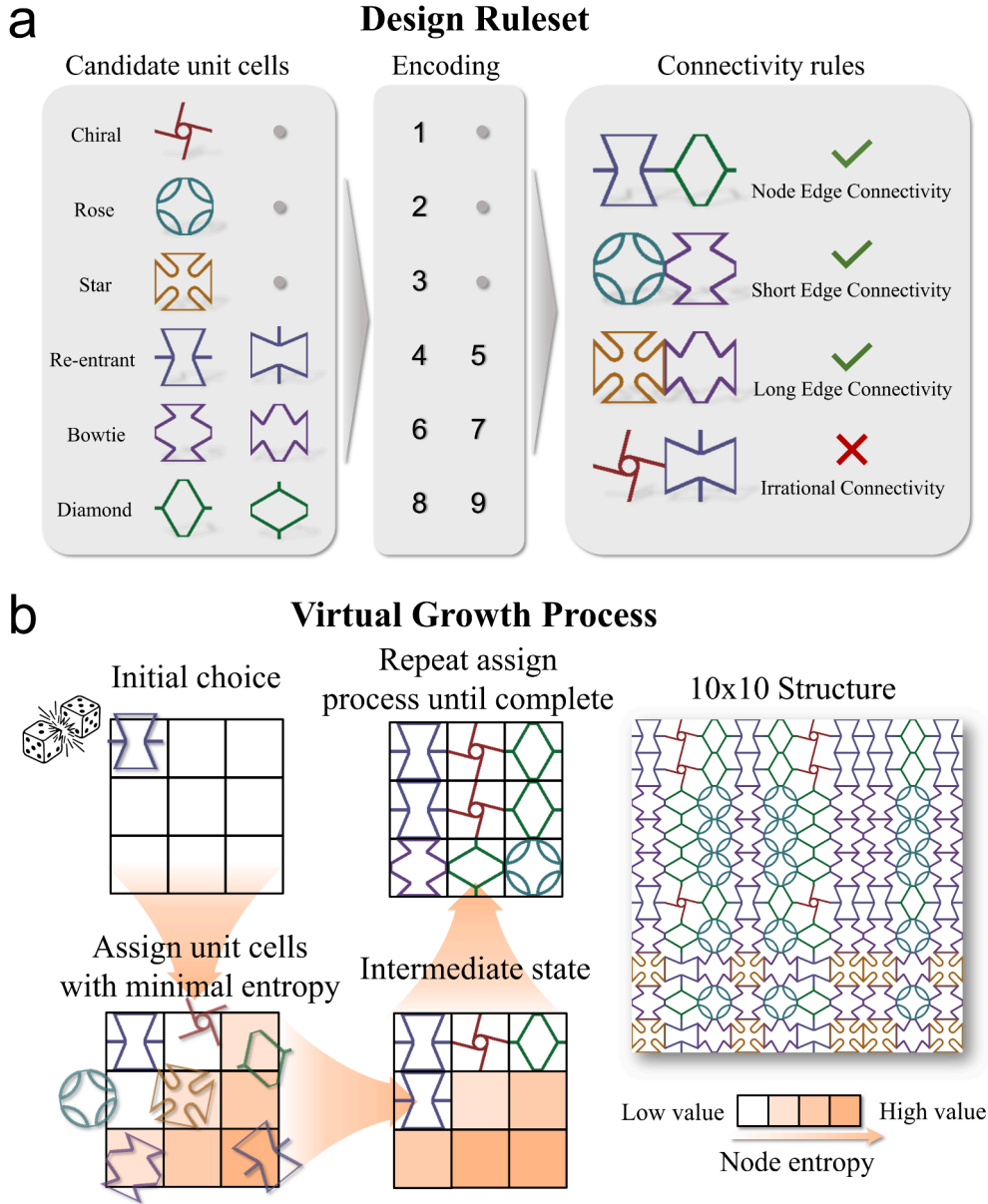


Fig. 2. Schematic of the VGA for generating metamaterial structures. (a) A library of candidate unit cells is established, with each unique cell encoded numerically. Predefined connectivity rules govern the valid connections between different cells to ensure connectivity, while prohibiting irrational connections. (b) The growth process begins with a random initial choice of a unit cell placed in the grid. Subsequent cells are iteratively assigned to positions with the minimal node entropy, representing the locations with the fewest placement options. This process is repeated until a complete 10 by 10 structure is generated.

This approach inevitably introduces boundary effects near the edges. However, the selected 10×10 lattice size provides a sufficient volume to ensure that the bulk response dominates the overall mechanical behavior. While the stress state of edge unit cells differs from that of the central ones, the global nonlinear responses remain consistent and representative at this scale [80]. Moreover, the data-driven framework implicitly integrates these boundary conditions into the learning process. Both the models capture the mapping between the disordered architecture and its global response including the boundary constraints.

Finite element analysis provides a controlled ground truth for assessing the inverse design framework, especially its ability to satisfy strict connectivity while matching target nonlinear responses. In practical realization, fabrication variability and testing uncertainty can influence the measured nonlinear stiffness response, so separating methodological validity from physical variability remains important. Future work extends the current framework toward physical evaluation to quantify the sensitivity of the designed responses to such

imperfections.

2.3. Forward prediction with Fourier neural operator

To evaluate the performance of the generated designs, a DL-based surrogate model is employed for forward prediction. A powerful paradigm is Operator Learning, which aims to learn the underlying operator that maps between infinite-dimensional function spaces [81–83]. As illustrated conceptually in Fig. 4a, two function spaces are considered, an input space A and an output space U , defined over a bounded domain D . The goal is to learn a nonlinear operator $G^\dagger$ that maps an input function a from to an output function v in U . Given a set of observed input-output data pairs, the aim is to approximate the true operator $G^\dagger$ with a parameterized model G_θ . This is formulated as an optimization problem where we find the optimal parameters θ that minimize the expected value of a cost function, which measures the difference

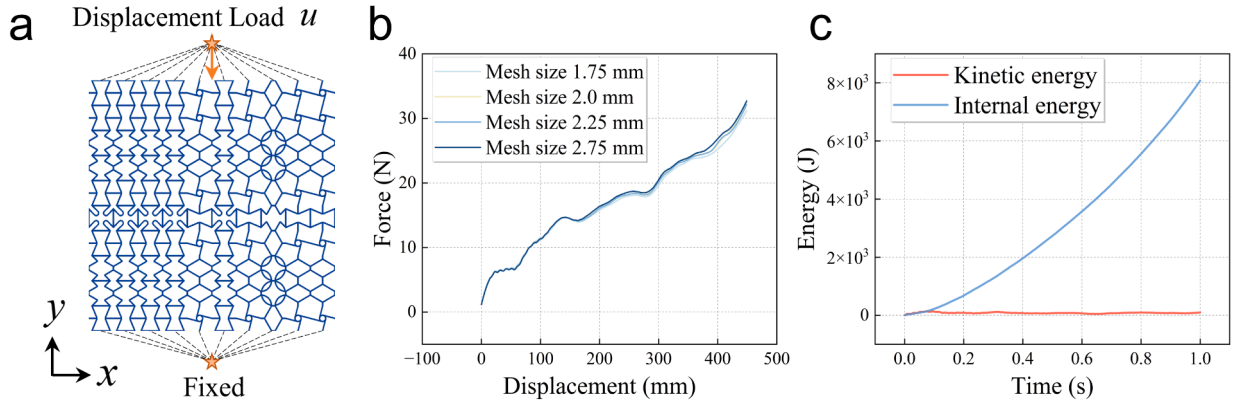


Fig. 3. Schematic of FE setup. (a) The FE simulation model setup, showing the metamaterial structure with applied displacement loading and fixed boundary conditions. (b) Mesh convergence study, showing that the force-displacement results are consistent across different mesh sizes. (c) Energy history plot, indicating a quasi-static loading condition, as the kinetic energy is negligible compared to the internal energy.

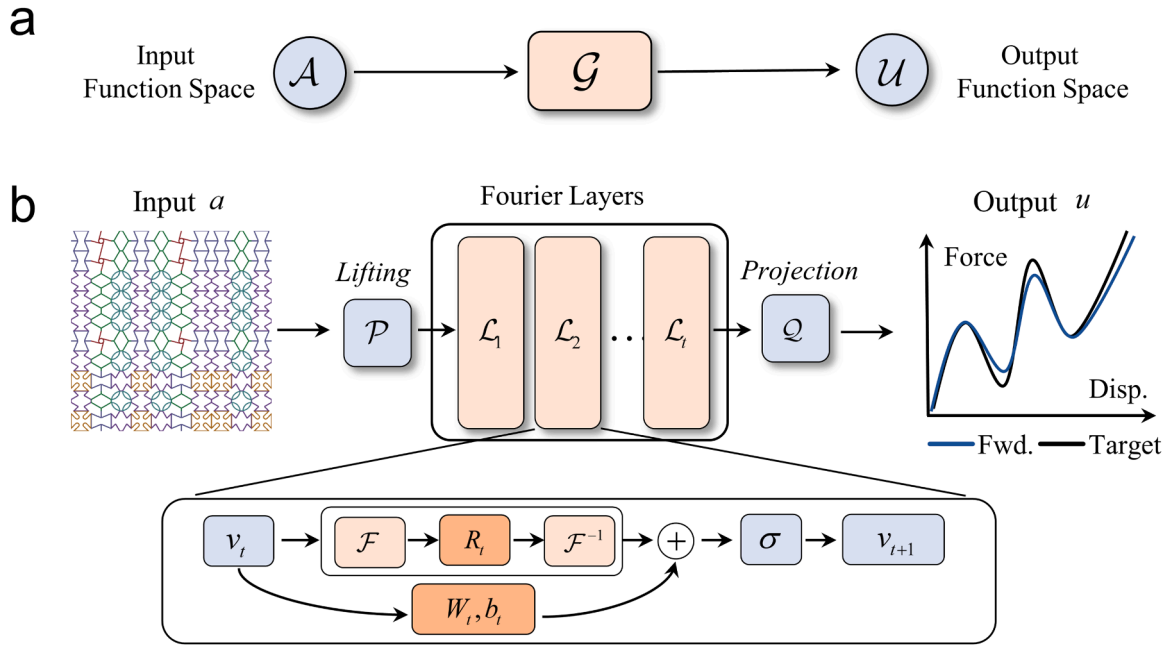


Fig. 4. Architecture of the FNO as the forward prediction surrogate model. (a) A conceptual schematic of Operator Learning, where it learns an operator G that maps A function a from the input space to a function U in the output space. (b) The detailed FNO architecture implemented in this work. The input metamaterial structure is first lifted to a high-dimensional representation by an encoder P . This representation is then processed through a series of Fourier layers, which efficiently learn the system's global features in the frequency domain. Finally, a decoder Q projects the data back to the target space to yield the predicted nonlinear response curve.

between the model's prediction and the ground-truth output. The process is expressed as follows:

$$\min \mathbb{E}[C(G_\theta(a), G^\dagger(a))] \quad (2)$$

The construction of G_θ can be realized through deep learning techniques. Among various methods, the neural operator framework is formulated as an iterative architecture, where each iteration updates the hidden representation v_{t+1} as follows:

$$v_{t+1}(x) = \sigma(W_t v_t(x) + (K(a; \phi) v_t)(x)) \quad (3)$$

where the process starts with an initial representation $v_0(x) = P(a)$, here P is a lifting operator. Each subsequent layer consists of a local linear transformation performed by a neural network W_t and a global integral operator K . σ is a nonlinear activation function. The final output is obtained by projecting the last layer's representation $u(x) = Q(v_{t+1})$, where Q is the projection operator.

In the intermediate iterative process, two key operators are used: a

linear local transformation W_t , and an integral operator K defined as:

$$(K(a; \phi_t) v_t)(x) := \int_D k(x, y, a(x), a(y); \phi_t) v_t(y) dy \quad (4)$$

where the kernel function k is parameterized by a set of learnable parameters ϕ_t , which can be realized with a neural network. The final solution u is obtained through a projection operator Q , which is typically realized with a neural network.

Specifically, the FNO is employed as an advanced surrogate model, which can provide fast and accurate performance evaluations [83,84]. The FNO obtains a specialized integral formula by imposing translation invariance on the kernel function and eliminating the dependency on a :

$$(K(a; \phi_t) v_t)(x) = \int_D k_{\phi_t}(x - y) v_t(y) dy \quad (5)$$

By applying the convolution theorem, the integral calculation can be simplified. Let F denote the Fourier transform operator acting on the

function v , and F^{-1} be its inverse transform. Let $R_t := F(k_{\phi_t})$ denote the Fourier transform of the kernel, then we can obtain:

$$(K(a; \phi_t) v_t)(x) = F^{-1}(R_t \cdot F(v_t))(x) \quad (6)$$

As shown in Fig. 4b, in this work, the input two-dimensional meta-material structure is treated as a function defined over a spatial domain, and the resulting nonlinear response curve as a target output function v . The architecture of the FNO consists of three main components. The process for each Fourier layer is expressed as:

$$v_{t+1}(x) = \sigma(W_t v_t(x) + F^{-1}(R_t \cdot F(v_t))(x)) \quad (7)$$

where the overall architecture is composed of an initial lifting operation $v_0(x) = P(a)$, followed by a series of t Fourier layers as defined in Eq. (7), and a final projection $u(x) = Q(v_{t+1})$. The operator W_t is a linear transformation, often a simple linear layer with bias b_t , and R_t is the learnable weight tensor in the Fourier domain for layer t .

2.4. Inverse design with autoregressive model

Our token-based AR model directly echoes the step-by-step, probabilistic decision-making of the VGA. While the VGA provides a foundation of valid geometric samples through stochastic assembly, it lacks the mathematical capacity to identify specific topologies that satisfy target mechanical requirements. The AR model serves as the inverse mapping operator that internalizes these geometric constraints while simultaneously optimizing the structural configuration to match the conditional prompt. At its core, an AR model operates on conditional probability, a process known as Causal Modeling [85]. This process can be formulated as:

$$p(x) = \prod_{i=1}^N p(x_i | x_1, \dots, x_{i-1}, c) \quad (8)$$

where the probability of the entire sequence of decisions, denoted by x , is calculated by multiplying all previously placed decisions (x_1 to x_{i-1}) and the overall condition c . This formulation ensures that the generation process is not a simple reproduction of connectivity but a goal-oriented search within the valid design space.

As shown in Fig. 5a, during the generation phase, the AR model takes a condition and makes a decision for the current step based on all the steps that came before it. To prepare the inputs, the target nonlinear response curve $l \in \mathbb{R}^{301}$ is projected into the model's hidden dimension $d_{hidden} \in \mathbb{R}^{512}$ using a Multi-Layer Perceptron (MLP) to act as the conditional prompt. The generation process is initiated by a learnable start token of the same dimension and proceeds in a raster scan order, from left to right and top to bottom. This sequence is then resized into the final 10 by 10 structure. All input embeddings are combined with absolute positional encodings to provide the model with the spatial information for this 2D layout.

For the detailed architecture, as shown in Fig. 5b, the implementation adopts the predominant decoder-only Transformer architecture, given its proven effectiveness as the foundation for state-of-the-art AR models. This approach is chosen for its attention mechanism, which can be visualized as attention maps to understand how the model weighs different previous decisions when making a new decision, preventing the model from being a complete black box. Furthermore, this architecture offers distinct advantages over other generative paradigms for this specific discrete assembly task. Unlike Diffusion models which operate in continuous space and struggle with strict discrete constraints, the Transformer natively manages categorical unit cell selection without post-processing. Compared to the complex reward engineering required in RL or the fixed-topology limitations of GNNs, the Transformer's causal attention mechanism efficiently learns both local compatibility and global dependencies through stable supervised training.

Specifically, the model employs a Causal Multi-Head Attention

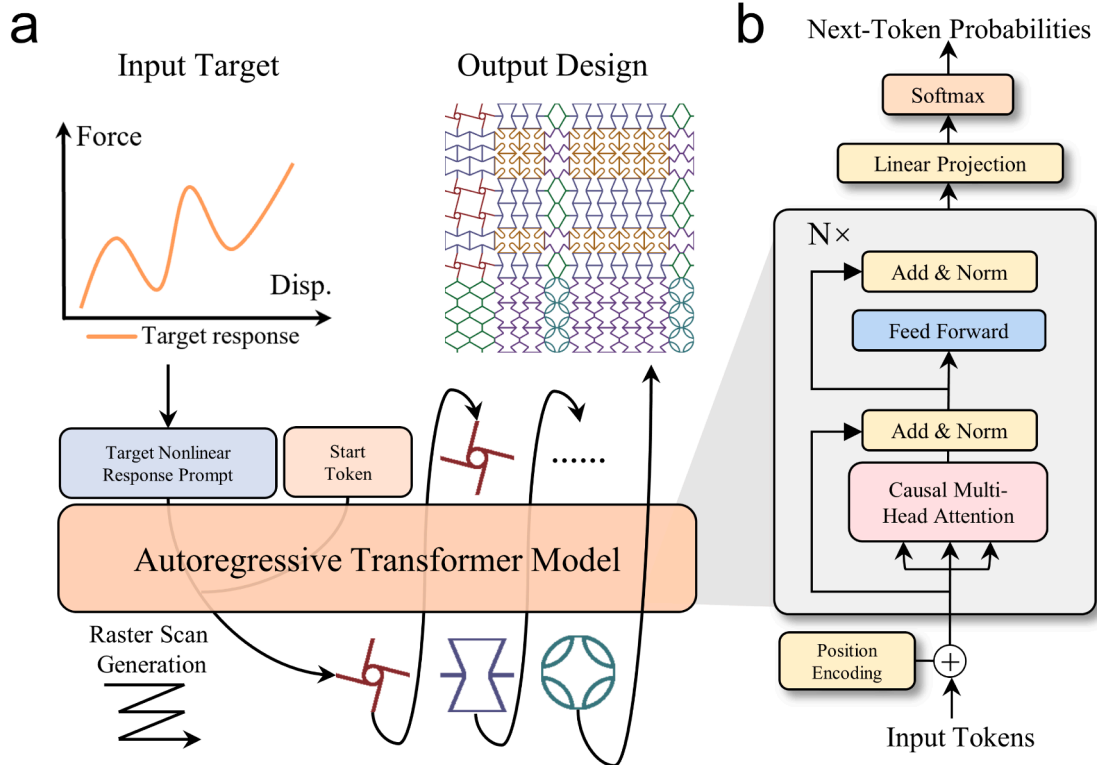


Fig. 5. The AR model for the inverse design of disordered metamaterials. (a) The overall inverse design workflow. The model takes the target nonlinear response as a conditional prompt and autoregressively generates the corresponding metamaterial structure. The generation proceeds token-by-token following a raster scan order, initiated by a start token. (b) The architecture of the Transformer block used in the model. At each step, the Causal Multi-Head Attention mechanism ensures that the prediction for the next unit cell is based solely on the previously generated cells and the target prompt, thereby learning the implicit rules of structural connectivity.

mechanism which can be formulated as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right)V \quad (9)$$

where Q seeks information, K is used for matching, and V is the information it provides. M is a matrix that forces the model to only pay attention to previous tokens in the sequence. $\sqrt{d_k}$ improves training stability by preventing the dot products from becoming too large.

During the training stage, the attention mechanism allows the use of a method known as teacher-forcing. In this approach, we always provide the ground-truth encodings of unit cells from the training data as the input for the next step. The model is trained to minimize the Cross-Entropy Loss between its predicted probability distribution and the ground-truth distribution for each position in the sequence. The loss function can be formulated as:

$$L = - \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(p_{ij}) \quad (10)$$

where N is the sequence length, which is 100 in this case, and C is the number of unique unit cells. For each position i , the term y_{ij} represents the ground-truth, which is a one-hot encoding indicating the correct unit cell. The term p_{ij} is the model's predicted probability for that same unit cell, generated by a softmax function over the model's output logits.

During the inference stage, Top-P Sampling is employed to balance creativity and accuracy in this process. This method first sorts the potential unit cells by their predicted probabilities in descending order. It then selects the smallest possible set of top candidates whose cumulative probability is greater than or equal to a predefined value P , which is set to 0.95 in this case. Temperature acts as the primary control parameter for the design trade-off, while Top-P functions as a foundational stability filter to facilitate random sampling of the final unit cell from the reduced set. Top-P sampling introduces a degree of randomness during the inference process. This mechanism provides a library of candidate designs rather than a unique solution. This diversity offers flexibility in engineering applications where specific layout requirements may differ. The probability distribution itself is generated by applying a softmax function to the model's output logits, which is modulated by a temperature parameter t . The softmax function can be formulated as:

$$p_i = \frac{\exp(z_i/t)}{\sum_{j=1}^C \exp(z_j/t)} \quad (11)$$

where p_i is the final probability for the i -th unit cell, z_i is the logit value for that cell, C is the total number of unit cell classes, and t is the temperature.

It is imperative to note that the model is explicitly trained to learn the specific sequential assembly patterns inherent in the VGA. The generation process avoids pre-defined masks or hard constraints intentionally, relying solely on the learned probability distribution to resolve connectivity. This strategy verifies that the autoregressive architecture effectively internalizes the complex geometric compatibility rules of the VGA data, demonstrating a sequential reasoning capability that distinguishes it from baseline models which fail to capture these strict dependencies.

The autoregressive architecture accommodates varying grid resolutions by adjusting the sequence length to match the structural scale. Furthermore, the conditional embedding mechanism extends to other physical scenarios, such as shear loading or multi-objective targets, by processing these requirements as numerical input vectors. While the current model parameters are specific to the defined material and loading conditions, the framework generalizes to different constitutive laws through retraining on corresponding datasets without architectural modification.

3. Results and discussion

3.1. Visualization and characterization of the nonlinear response dataset

Fig. 6 displays the nonlinear force-displacement response curve for the disordered architected metamaterial under quasi-static compression. This behavior arises purely from geometric nonlinearities because the base material is hyperelastic. The entire deformation occurs within the elastic region without plastic yielding. Given that the focus of this study is on the geometric nonlinearity derived from the structural architecture, material damage and fracture mechanisms are not accounted for in the current finite element simulations. Specifically, initially the structure exhibits a linear elastic response as high stiffness unit cells like Star or Bowtie compress. The first plateau region begins when certain slender members within unit cells like Chiral or Re-entrant experience elastic buckling. Crucially, the target nonlinear response relies on such controlled local instabilities. The autoregressive model implicitly learns to arrange slender members to trigger specific deformation modes that enhance energy absorption, treating local instability as a functional mechanism rather than a failure mode. Stiffness then increases again as deformation progresses and internal structural members initiate self-contact. Subsequent plateaus occur as other sections of the structure undergo similar elastic buckling events. The increasing stiffness between plateaus is dominated by accumulating self-contact. This sequential pattern of elastic buckling and self-contact creates the complex nonlinear curve. The specific shape is a direct result of the unique spatial arrangement of different unit cells within the disordered cellular architecture.

Following the FE simulation previously stated, a comprehensive dataset is generated comprising 2423 unique disordered metamaterial structures and their corresponding nonlinear compressive response curves. Notably, different training and validation sets are prepared for the inverse design and forward prediction tasks.

For the inverse design task, we first visualize the data distribution by applying Uniform Manifold Approximation and Projection (UMAP) to reduce the dimensionality of all response curves to only two dimensions [86]. A k-means algorithm is employed to group the dimensionless data into four clusters and identify their corresponding centroids [87]. As shown in Fig. 7a, a representative validation set is created by randomly sampling 50 curves from each cluster, totaling 200 samples. These samples are strictly excluded from the training set to prevent data leakage, ensuring that the inverse design performance is evaluated on completely unseen target responses. The uniform distribution of these validation samples within each cluster confirms that our subset is representative of the overall data diversity. To further validate the clustering, the response curves corresponding to the four cluster centroids are visualized in Fig. 7b. The colors of these curves correspond to their originating clusters. A progression is evident across these representative responses, moving from Centroid 1 to Centroid 4, there is an increase in the maximum reaction force. The complexity of the nonlinear responses, characterized by the number and magnitude of peaks and valleys, also progressively increases.

For the forward prediction task, the full dataset is partitioned using 8/2 split for training and validation. The UMAP visualization in Fig. 7c shows that the validation samples are evenly distributed among the training data, showing a representative and reasonable split. Notably, the validation set for the forward model is constructed to contain the validation set for the inverse design model.

Fig. 7d provides an overview of all force-displacement curves. The data from the training set establishes a *Training Set Bound*, which defines the known design space and represents the theoretical performance limits for interpolation tasks. The dense and uniform distribution of curves within this bound suggests that the nonlinear response design space is well-covered. To evaluate the model's extrapolation capabilities, response curves are also generated from periodic structures, which yield the minimum and maximum strength responses. These curves

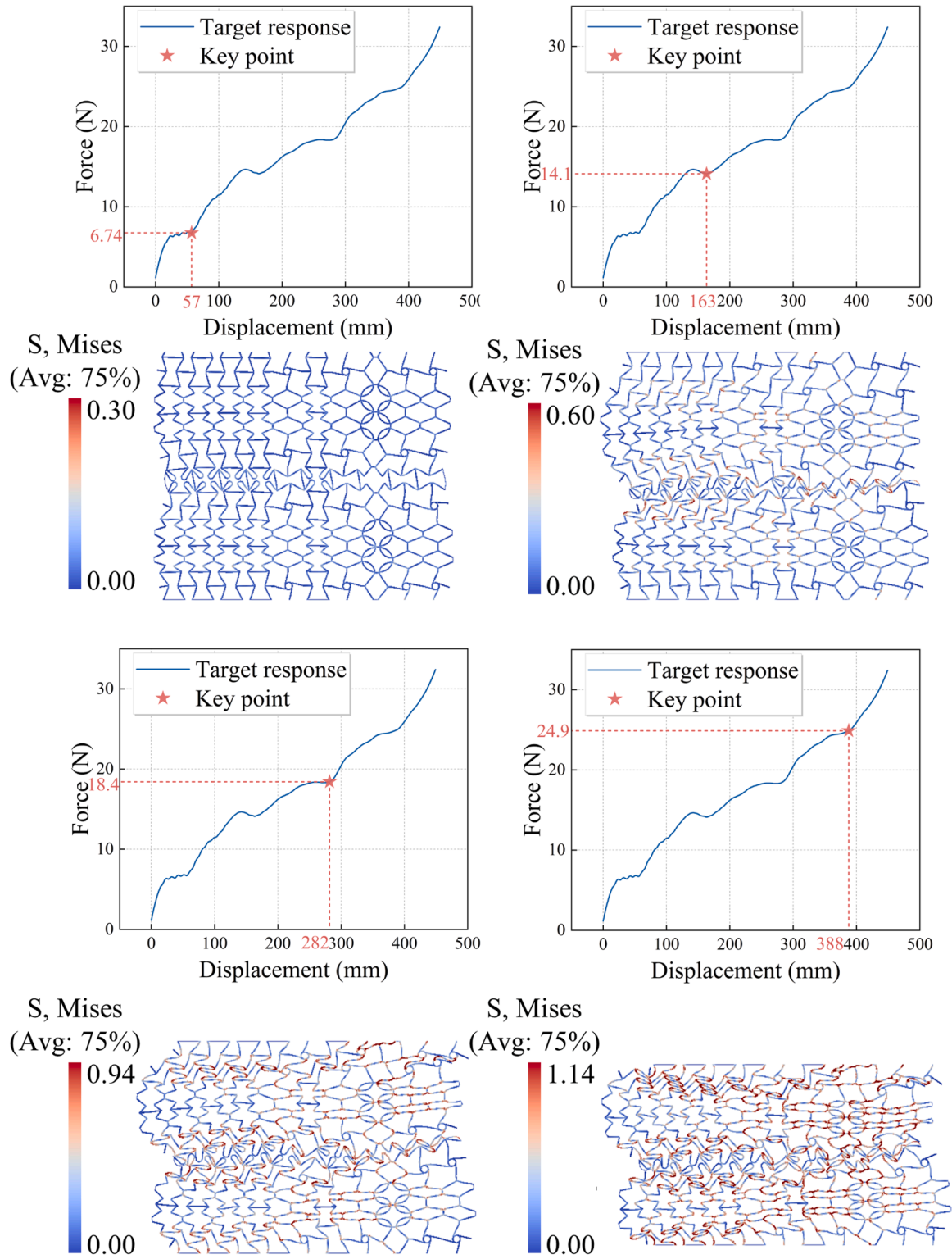


Fig. 6. Verification and analysis of FE simulation results. The progressive deformation sequence under compression. The top row shows the force-displacement response with key points highlighted, while the bottom row displays the corresponding von Mises stress distribution in the structure at each of these points.

define an *Ideal Design Bound* situated outside our training data, indicating that there is still some extrapolation potential for the design space beyond the training bound.

3.2. Forward prediction performance: evaluation of the FNO surrogate model

In this section, the forward prediction performance of the FNO is evaluated. The assessment is conducted from two perspectives: an overall statistical comparison and a detailed case-study analysis. The FNO is benchmarked against a conventional Unet architecture with a

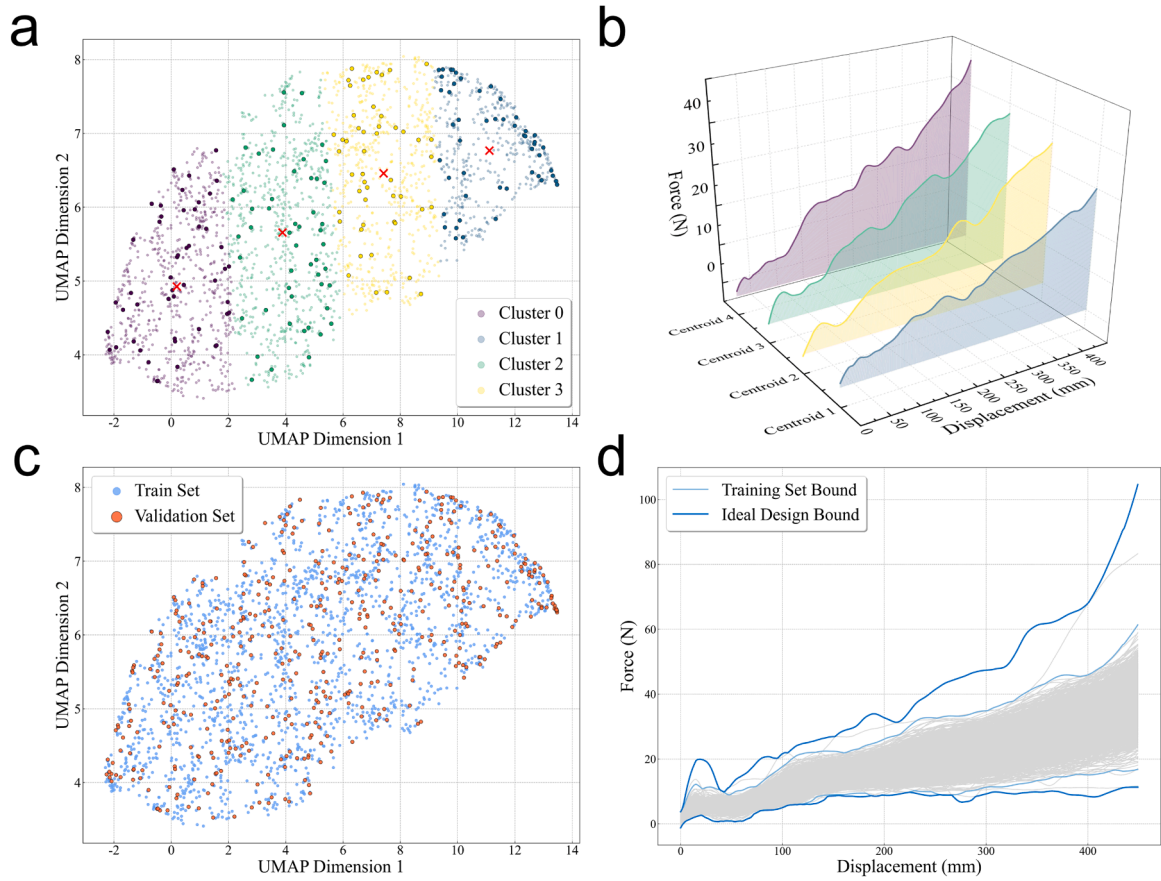


Fig. 7. Visualization of the datasets for forward prediction and inverse design. (a) UMAP visualization of the inverse design dataset, showing four k-means clusters. The dataset is split into a lowlight training set and a highlight validation set. (b) The four representative response curves of the cluster centroids. (c) UMAP visualization of the forward prediction dataset, showing the train/validation split. (d) The curves from the training set are shown in grey, defining the model's known data distribution. The *Ideal Design Bound* represents a target performance space intentionally set outside of the training set to evaluate the model's generalization capabilities.

similar number of parameters [88], to show the effectiveness of the Operator Learning paradigm for our specific application.

To quantitatively measure the accuracy of the predictions, three metrics are employed: the Mean Relative Error (MRE), the coefficient of determination (R^2), and the Normalized Root Mean Square Error (NRMSE). While MRE and R^2 are used for the overall performance assessment, NRMSE serves as the main indicator for evaluating and

ranking the case-study results. The mathematical formulations for these metrics are defined as follows:

$$MRE = \frac{1}{n} \sum_{i=1}^n \frac{y_{pred,i} - y_{true,i}}{y_{true,i}} \times 100\% \quad (12)$$

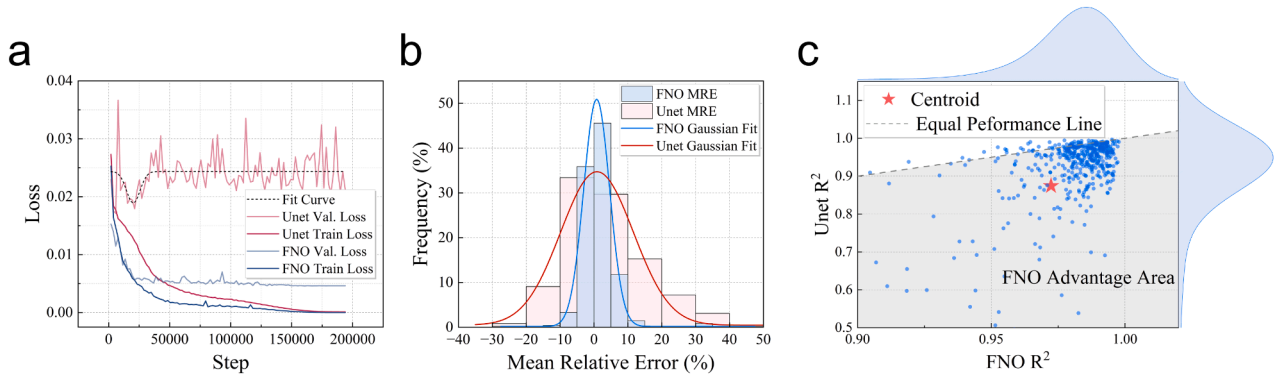


Fig. 8. Overall comparison of forward prediction results between the FNO and Unet models. (a) Training and validation loss curves for FNO and Unet with a similar number of parameters. The FNO model shows stable convergence, whereas the Unet model shows clear signs of overfitting, as confirmed by the fitted curve of its validation loss. (b) A histogram comparing the distribution of MRE for both models. (c) A scatter plot comparing the coefficient of determination R^2 for each sample. The plot includes an equal performance line and a centroid representing the mean R^2 values. Most data points are located in the region defined as the FNO Advantage Area.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_{pred,i} - y_{true,i})^2}{\sum_{i=1}^n (y_{true,i} - \bar{y}_{true})^2} \quad (13)$$

$$NRMSE = \frac{\sqrt{\sum_{i=1}^n (y_{pred,i} - y_{true,i})^2}}{\sqrt{\sum_{i=1}^n y_{true,i}^2}} \quad (14)$$

where y_{true} represents the ground-truth response curve from the FE simulation, y_{pred} is the curve predicted by the surrogate model, $\bar{y}_{true}$ is the mean of the ground-truth values, and n is the number of discrete points along the curve which is 301 in our case.

At the overall level, the performance of the two models is first assessed by examining their loss functions, as shown in Fig. 8a. The FNO model contains 3.1 M trainable parameters, whereas the Unet model has 5.7 M. The Unet architecture is more susceptible to overfitting when training samples are limited, which results in the observed rebound in validation loss. Conversely, the FNO model utilizes the operator learning paradigm to learn the underlying mechanical response as a mapping between function spaces. This characteristic provides superior generalization capabilities across diverse metamaterial configurations even with a small dataset. Consequently, the FNO model trains effectively and

achieves stable convergence with a minimal gap between the final training and validation loss values. Next, the prediction accuracy is compared using the MRE, as depicted in the histogram in Fig. 8b. The distribution of MRE for the Unet is more dispersed, with a flatter Gaussian fit and a number of predictions in the tails of the distribution. Conversely, the MRE for the FNO is concentrated within the -10% to 10% range, resulting in a much sharper Gaussian peak. Finally, the scatter plot in Fig. 8c compares R^2 for each prediction from both models. For visual clarity, some extreme outliers from the Unet have been omitted from the plot. The *Equal Performance Line* indicates where both models would have identical accuracy. The vast majority of data points fall within the FNO advantage area, showing that FNO outperforms Unet on a sample-by-sample basis. To quantify this, the centroid of the entire data distribution is calculated, including all outliers. The centroid's position reveals that the mean R^2 for FNO is in the 0.95 to 1.0 range, while the mean R^2 for Unet is in the 0.8 and 0.9 range, which further shows the greater predictive performance of FNO overall.

At the case-study level, the NRMSE is used as the main performance metric. This value quantifies the similarity between the predicted curve and the ground-truth curve, where a higher NRMSE indicates a larger discrepancy. In the plots, the shaded region, referred to as the uncertain area, visually represents the magnitude of this prediction error. To provide a comprehensive assessment, three representative cases are

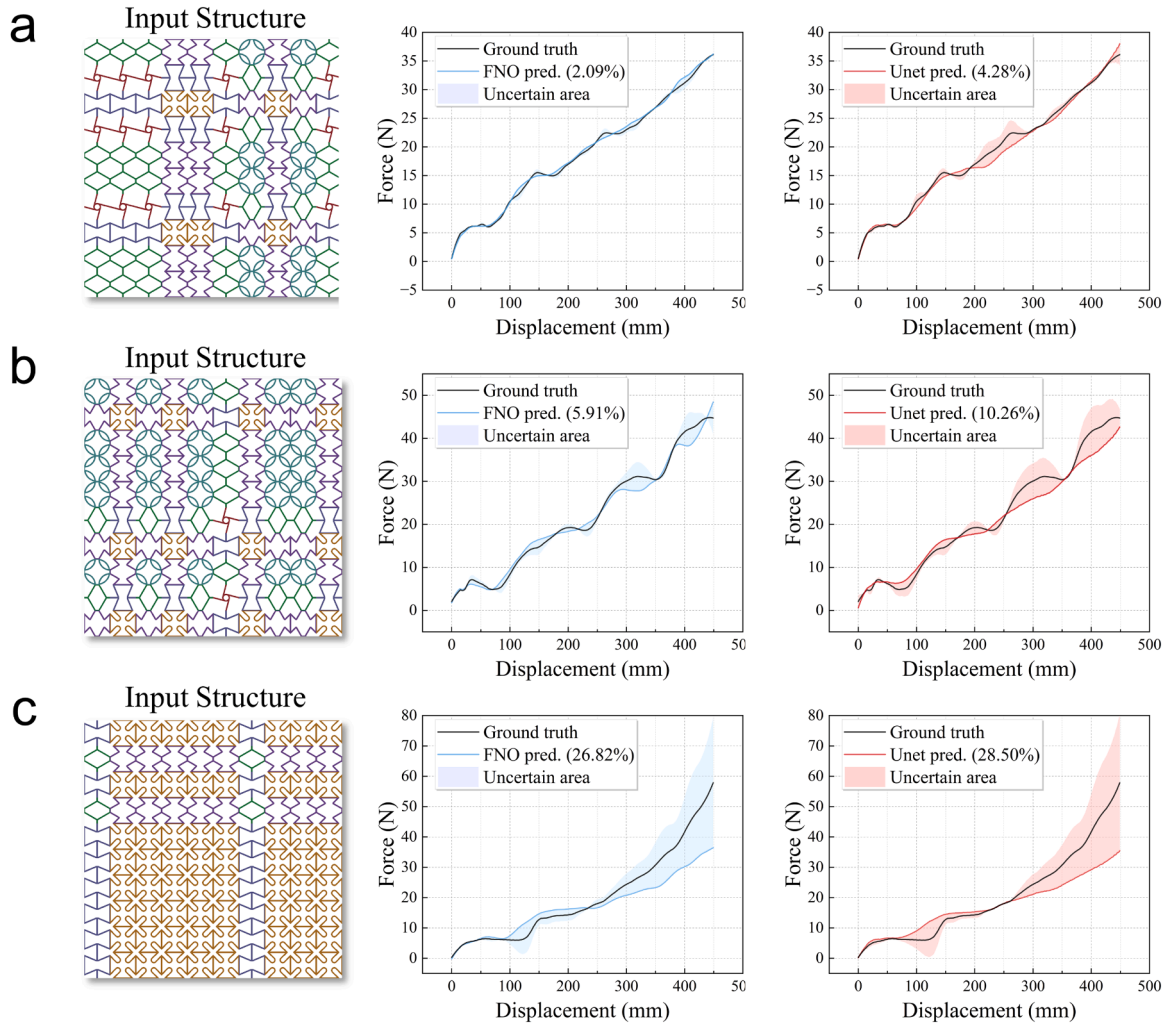


Fig. 9. Case-study comparison of forward prediction results between the FNO and Unet models. Three representative cases are selected from the test set to illustrate the spectrum of FNO's prediction performance, with the corresponding Unet prediction for each case shown for comparison. (a) The best-case prediction for FNO, with the lowest NRMSE. (b) The median-case prediction for FNO, selected based on the median NRMSE. (c) The worst-case prediction for FNO, with the highest NRMSE. Legend percentages represent NRMSE values, while the shaded area shows the prediction uncertainty.

selected from the validation set that correspond to the best, median, and worst NRMSE scores achieved by the FNO model. Fig. 9a displays the best-case prediction from the FNO model. The predicted curve aligns almost perfectly with the ground-truth, accurately capturing the three distinct local peak and valley features characteristic of this structure's nonlinear response. In contrast, the Unet model, despite achieving a relatively low error, predicts the second valley prematurely and fails to capture the third valley, resulting in lower accuracy. A more typical result is presented in Fig. 9b, which shows the median-NRMSE case for the FNO. While the precision is lower than in the best case, the FNO successfully outlines the overall trend and the maximum force of the response curve. However, it incorrectly predicts the position of one valley and introduces an additional valley. The Unet's performance on this sample is worse. The most challenging example is shown in Fig. 9c, representing the worst-case prediction for the FNO. Here, neither model accurately predicts the exponential-like nonlinear response, though the FNO's prediction remains closer to the ground-truth than the Unet's. An explanation for this poor performance is that the input structure closely resembles a periodic design. Such structures are rare in our training dataset of disordered metamaterials. Furthermore, the response curve's maximum force exceeds the range of the training data, requiring the models to extrapolate beyond their learned distribution.

3.3. Capabilities of the autoregressive inverse design framework

3.3.1. Interpreting the sequential generation mechanism

Before diving into the inverse design results, it is crucial to first examine the model's decision-making process. Fig. 10 illustrates how the model generates a structure when given a target nonlinear response. In Fig. 10a, the *reference structure* is defined as the structure whose simulated compressive response curve is used as the design target. It is important to clarify that the primary objective is to minimize the deviation between the generated design's response and the target response. The geometry of the output structure does not need to be identical to the reference structure. A design is also considered successful if it produces a

novel structure whose mechanical response still falls within an acceptable tolerance of the target.

At the macro level in Fig. 10b, the performance evaluation confirms the success: the design's response closely matches the target curve with only minor discrepancies, resulting in a very small uncertain area. Simultaneously, at the micro level, the decision process has also been visualized at several key locations in Fig. 10b. For Token 2, the position requires a node-edge connectivity at the beginning of the sequence. The model correctly identifies that only Chiral, Re-entrant, and Diamond units are valid choices, assigning near-zero probability to all other options. It then selects the most probable option, Re-entrant, ensuring connectivity from the start. Token 51 is at a critical location that influences both initial load-bearing and later stiffness increase induced by self-contact. Furthermore, the position requires a long-edge connectivity. The model again correctly identifies the three valid unit cells but chooses the Star. The probability distribution at Token 51 is notably dispersed compared to the deterministic predictions at the sequence boundaries. This dispersion occurs because the central region offers maximum topological freedom; multiple unit types, including the Star, Re-entrant, and Diamond, can satisfy local connectivity while providing valid stiffness contributions for the target response. Consequently, the model assigns comparable probabilities to these options, reflecting the existence of divergent but equally viable design paths. In contrast, as the sequence progresses to Token 100, the accumulation of geometric constraints from fully defined neighbors collapses the design space. The Rose unit becomes the sole topological solution satisfying the strict boundary connectivity, which the model captures by assigning a probability approaching 100% to this single valid choice.

The probability assignments originate directly from the softmax output of the decoder, conditioned on both the target response and the preceding structural sequence. The occurrence of similarly high probabilities for multiple unit cells reflects mechanical equivalency, where distinct topological forms provide comparable stiffness or deformation characteristics at a specific location. Consequently, the proposed methodology prioritizes the convergence of the global mechanical

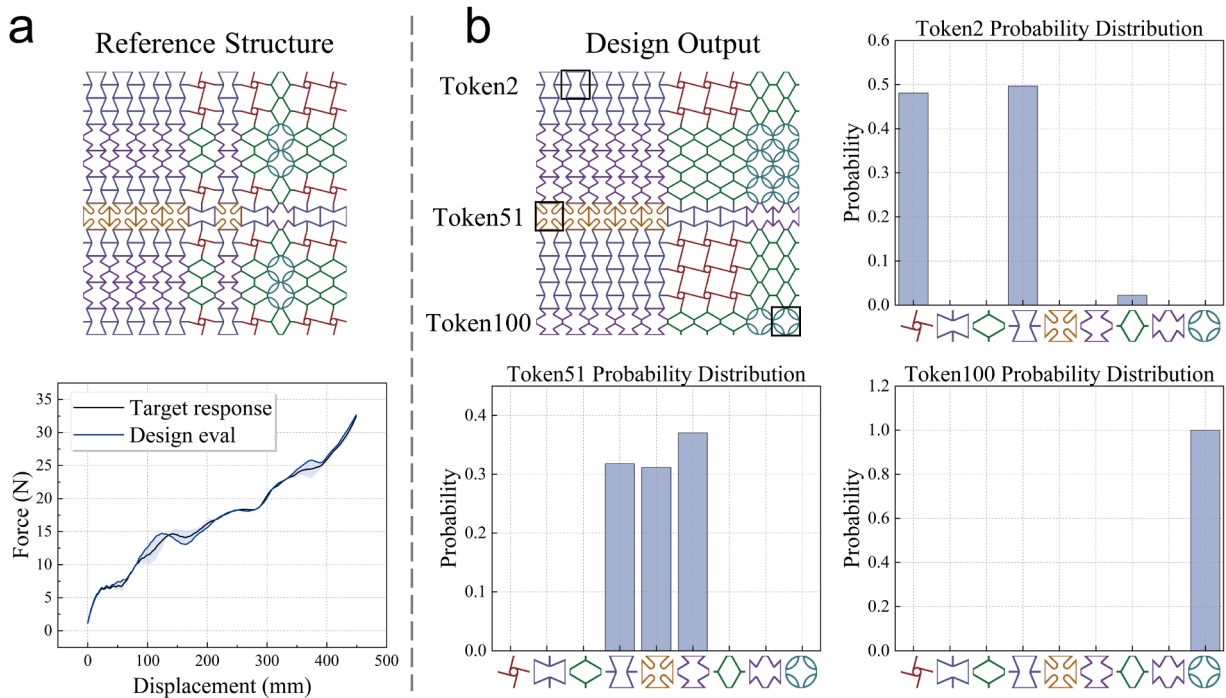


Fig. 10. Visualization of the decision-making process of the AR inverse design model. (a) A reference structure is shown with its corresponding target response curve. The plot also includes the evaluated response of the generated design, showing a strong agreement with the target. (b) The design output generated by the model. To illustrate the token-by-token generation, the probability distributions for the unit cell selection at the beginning (Token 2), middle (Token 51), and end (Token 100) of the sequence are displayed, revealing the model's decision process at each step.

response over the uniqueness of the geometric configuration. This approach inherently accommodates structural diversity, ensuring that while the resulting architectures may vary, their nonlinear performances consistently converge to the specified target.

To substantiate this implicit learning, the Appendix provides a detailed analysis of the attention mechanism. The attention maps reveal that the model spontaneously develops specific focus patterns on spatially adjacent neighbors to enforce local geometric compatibility. In summary, this process, combined with the attention map analysis, confirms that the model has learned implicit structural rules. Furthermore, the model ensures mechanical robustness through data-driven learning. By training on FEA-validated structures, it captures the implicit mapping between structural layout and stress distribution. Consequently, generated designs inherently favor reasonable load paths, tending to avoid severe stress concentrations that deviate from the valid physical distribution.

3.3.2. Trade-off between target performance and connectivity

This section investigates a critical trade-off between achieving the target performance and maintaining connectivity, a balance governed by the temperature parameter t in the generation process. It is noted that although the training dataset consists of valid structures, the generative model must independently learn the assembly rules to maintain integrity during the design of new configurations. As the temperature increases, the model explores more diverse unit cell arrangements to match the target nonlinear response. This improved performance is often accompanied by a higher incidence of connectivity errors. This relationship allows for the selection of a design strategy that balances structural integrity with the pursuit of optimal mechanical performance.

To quantitatively evaluate this trade-off, three normalized metrics are defined: the Connectivity Score, the Optimal Performance (OP) Score, and the Better-Than-Best (BTB) Score. The Connectivity Score measures the number of fully connected designs generated at a specific temperature. The OP Score counts the designs at a given temperature that achieve the lowest NRMSE among all designs generated across the entire temperature range for a single target. The BTB score counts the designs whose response is a closer match to the target than the best-matching example available in the training set. The Connectivity Score is an indicator of structural reliability, while the OP and BTB Scores serve as metrics for performance accuracy. The mathematical definitions are as follows:

$$\text{ConnectivityScore}(t) = \frac{|\{d \in D_t | d \text{ is fully connected}\}|}{|D_t|} \quad (15)$$

$$\text{OPScore}(t) = \frac{|\{d \in D_t | \text{NRMSE}(d) = \min_{d' \in D_t} (\text{NRMSE}(d'))\}|}{|D_t|} \quad (16)$$

$$\text{BTBScore}(t) = \frac{|\{d \in D_t | \text{NRMSE}(d) < \text{NRMSE}(d_{\text{match}})\}|}{|D_t|} \quad (17)$$

where t is the temperature, D_t is the set of all designs generated at this temperature, and $|D_t|$ is its total count. The term d represents a single design from the set D_t . The reference design d_{match} is the structure within the training dataset that has the lowest NRMSE to the target curve.

Fig. 11a presents a statistical analysis of how the generation temperature affects structural connectivity. Both excessively low temperatures, such as $t = 0$, and high temperatures, such as $t > 1.75$, produce a number of designs with connectivity problems. There is a viable range between $t = 0.25$ and $t = 1.5$ where the model reliably generates structures with few connectivity errors. Consequently, the subsequent performance evaluations focus exclusively on designs generated within these six temperature settings. The specific trade-off between connectivity and performance is illustrated in Fig. 11b. The radar chart reveals an inverse relationship. The Connectivity Score generally trends opposite to the OP and BTB Scores, indicating that maximizing structural integrity and achieving optimal performance are conflicting objectives. When the generation process is tuned to produce more designs with valid connectivity, the number of high-performance candidates decreases. Conversely, prioritizing performance leads to a less controlled design process where connectivity problems become more prominent.

Fig. 11c confirms this from the NRMSE distribution. Low temperatures $t = 0.25$ produce highly concentrated, mediocre designs. As temperature rises, the distribution flattens, increasing variance with both more high-NRMSE and low-NRMSE results, with this high-reward exploration peaking at $t = 1.25$. Quantitatively, the mean NRMSE across the validation set remains consistent, fluctuating reasonably between 8.02% and 8.71% across all temperature settings. The model achieves its highest precision at $t = 0.5$, recording a minimum NRMSE of 2.34%. However, the variance of the error distribution expands as the temperature rises; the worst-case NRMSE increases from 15.58% at the conservative setting of $t = 0.25$ to 29.82% at the more exploratory setting of $t = 1.25$.

3.3.3. Benchmarking structural connectivity against baseline models

To evaluate the capability of the generative models in maintaining structural integrity, a quantitative metric named the connectivity rate ζ is introduced. This metric represents the probability that a generated design satisfies all constraints required for physical feasibility. Mathematically, for a given set of generated structures, ζ is defined as the ratio of the number of fully connected samples to the total number of generated samples:

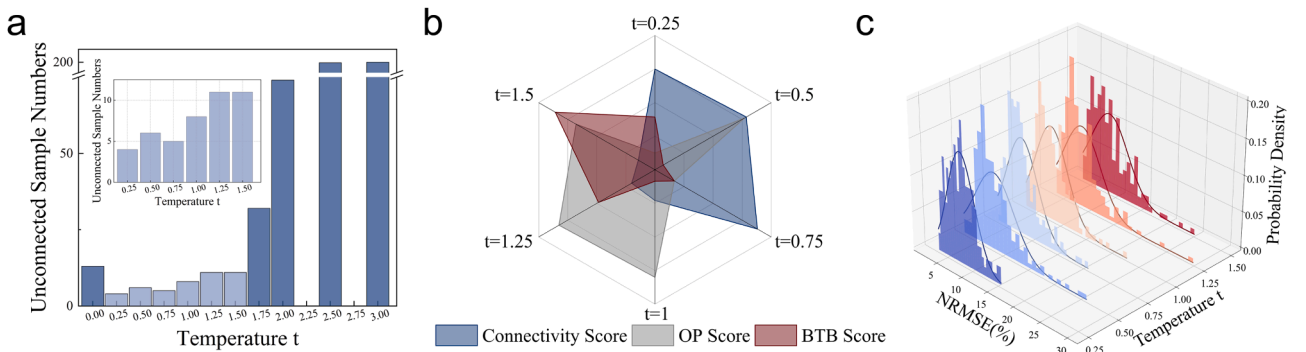


Fig. 11. Analysis of the trade-off between target performance and connectivity controlled by the generation temperature. (a) A bar chart quantifies the number of generated samples with connectivity errors. This plot is used to identify a viable range of temperatures where the model reliably produces structurally sound designs with a low failure rate. (b) A radar chart illustrates how higher performance metrics, represented by the OP Score and BTB Score, are achieved at the expense of the Connectivity Score. (c) A three-dimensional plot displays the probability distributions of the NRMSE, also viewed as an overall performance comparison of inverse design results.

$$\zeta = \frac{N_{fc}}{N_{total}} \quad (18)$$

where N_{fc} denotes the count of structures that strictly adhere to the connectivity rules without any isolated unit cells, and N_{total} represents the total size of the generated ensemble.

To validate the superiority of the proposed framework in maintaining structural integrity, we compare the connectivity rate distributions against several baseline models as illustrated in Fig. 12a. The Transformer model achieves a mean connectivity rate approaching 1.0 and outperforms all other methods. The Recurrent Neural Network (RNN) yields a mean rate of 0.98 but suffers from the vanishing gradient problem which limits its ability to capture long-term dependencies in complex sequences. The non-autoregressive models including Generative Adversarial Networks (GAN), Conditional Variational Autoencoders (cVAE), and Multi-Layer Perceptrons (MLP) exhibit inferior performance with mean rates of 0.91, 0.87, and 0.81 respectively.

Beyond the comparison with baselines, we further investigate the internal mechanisms of the Transformer by analyzing the impact of sampling temperature on connectivity stability. As shown in Fig. 12b, lower temperature settings consistently yield connectivity rates near 1.0. An increase in temperature flattens the probability distribution to facilitate exploration, which introduces a trade-off between structural diversity and connectivity adherence.

To specifically evaluate the severity of connectivity violations, we analyze the variations of mean, median, and worst-case connectivity rates as presented in Fig. 12c. The data reveals that the median connectivity remains consistently at the perfect level of 1.0 across the viable temperature range, confirming that the vast majority of generated designs are fully connected. Although the worst-case connectivity exhibits fluctuations and a downward trend at higher temperatures, these valid but imperfect designs represent rare outliers in the long tail of the distribution rather than a systemic failure. This analysis demonstrates that the model is robust against severe connectivity violations and that isolated defects can be effectively mitigated by selecting an appropriate temperature range.

3.3.4. In-distribution performance for inverse design

In this section, an interpolation case-study of the data-driven design framework's performance on the validation set is now presented. Representative best, median, and worst-case results are still selected to evaluate the model's capabilities. To quantitatively evaluate the fidelity of the inverse design, we employ the NRMSE between the target response and the simulated response of the generated structure.

First, the globally optimal design result is shown in Fig. 13a. Structurally, the optimal design output is highly similar to the reference

structure in its unit cell composition and arrangement, differing only by the inclusion of several Rose units at its base. The corresponding response curve shows exceptional performance, achieving a low NRMSE of 2.34%. It accurately captures the peaks and valleys of the target response and matches the maximum force almost perfectly. Furthermore, this design's performance exceeds that of the best-matching sample from the training set, proving the AR model's interpolation capabilities. The FNO prediction for this structure is also within an acceptable range of accuracy.

Next, Fig. 13b shows a median-case, which we use to illustrate the protocol for handling flawed connectivity generations. The phenomenon where a design with connectivity defects achieves a lower NRMSE arises because the global force-displacement response acts as an integral property, which tolerates minor local inconsistencies. However, such defects introduce local stress concentrations that render the structure physically inviable for practical engineering. Consequently, the proposed framework prioritizes structural integrity over marginal gains in curve-fitting accuracy. The selection of the fully connected, albeit sub-optimal, design in Fig. 13b represents a deliberate trade-off to ensure the connectivity, resulting in a slightly higher NRMSE of 4.33% compared to the disconnected optimal solution. To further mitigate the occurrence of such defects, the current framework utilizes batch sampling to generate a pool of candidates, ensuring the availability of valid solutions. Future iterations plan to integrate beam search algorithms to actively identify and correct topological violations during the sequence generation process.

Finally, the most challenging case is presented in Fig. 13c. The worst design output of this case exposes an inherent limitation of the AR paradigm: error accumulation. The performance of the design is characterized by a sudden drop in stiffness, indicating large scale buckling. This failure is a direct result of poor decisions made early in the generation process. The model places weak unit cells in the upper rows, which then constrains all subsequent choices. Because an AR model generates sequentially, with each step conditioned on all previous ones, an initial mistake can trigger a cascade of compromised decisions. The model is forced to make increasingly poor choices to satisfy local connectivity rules, leading to a deviation from the global performance target with an NRMSE of 25.97%. The inherent phenomenon of error accumulation in autoregressive generation can be mitigated through the introduction of additional inference strategies. Future work can integrate established Beam Search or Re-ranking techniques to optimize the design output. These methods allow the model to explore multiple candidate paths during the generation process, correcting early sub-optimal decisions by evaluating their impact on the global performance.

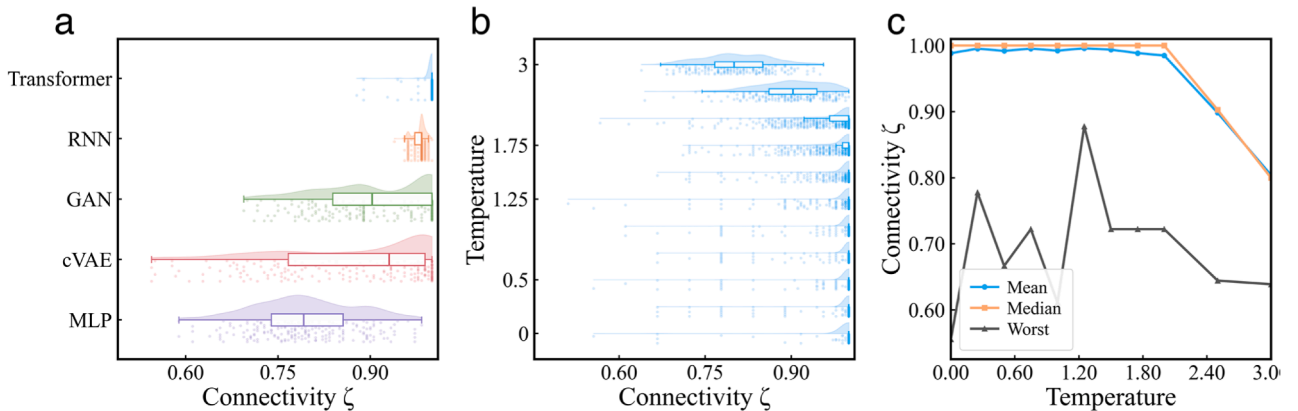


Fig. 12. Quantitative assessment of structural connectivity. (a) Comparison of the connectivity rate ζ distributions between the proposed Transformer model and baseline models, including RNN, GAN, cVAE, and MLP. (b) The distribution of connectivity rates for the Transformer model across varying sampling temperatures ranging from 0 to 3. (c) The variation of mean, median, and worst-case connectivity rates as a function of the sampling temperature.

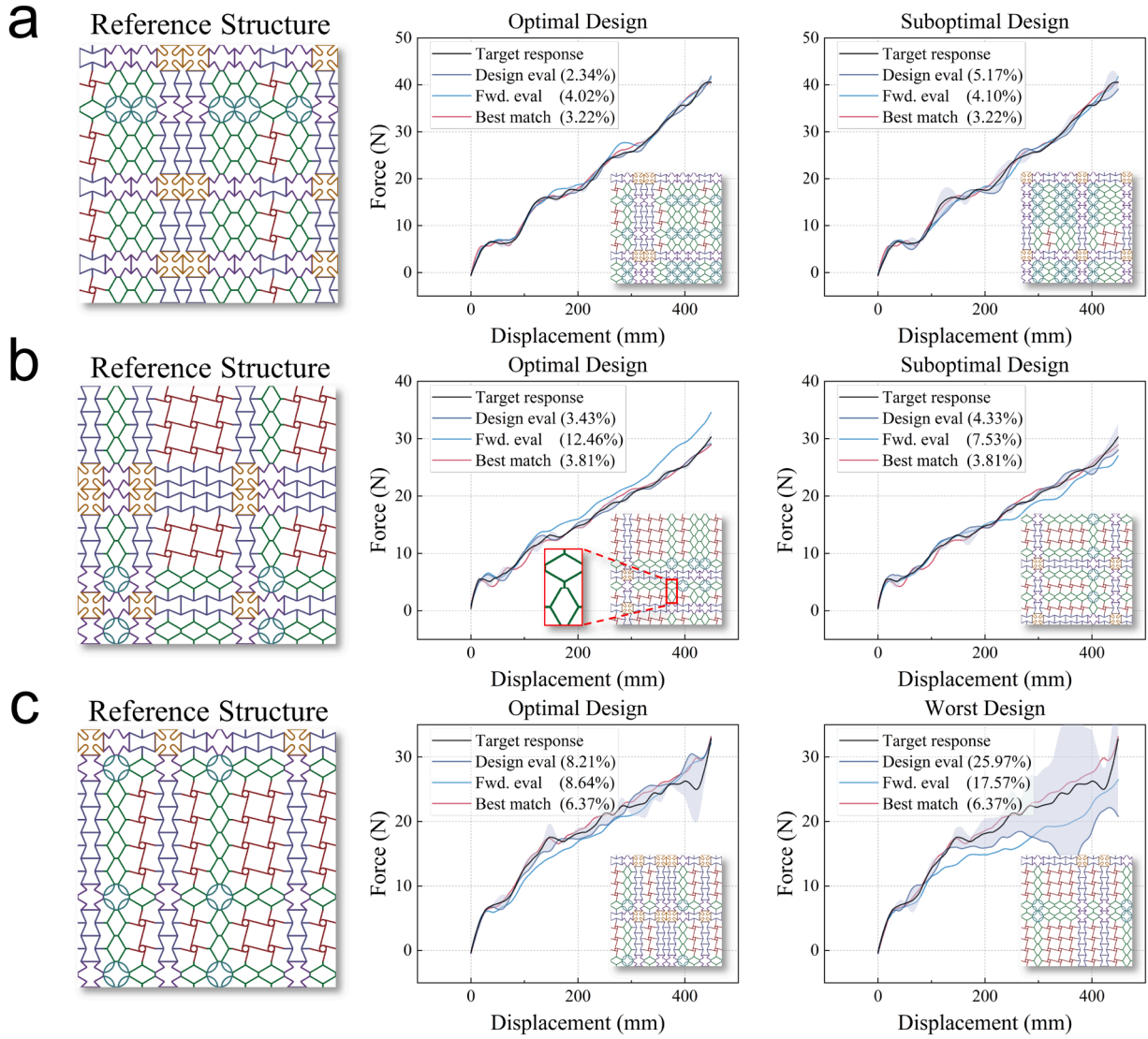


Fig. 13. Case-study of inverse design interpolation performance for optimal, median, and worst cases. (a) The globally optimal case, which achieved the lowest NRMSE in the validation set, is presented with both its optimal and suboptimal designs. (b) A median-NRMSE case is specifically chosen to show a strategy for handling generation flaws. Here, the optimal design contains a highlighted connectivity error. (c) The worst-NRMSE case is shown with its optimal and worst design results to show the performance range on a challenging target. Legend percentages represent NRMSE values, while the shaded area shows the deviation between the evaluated design response and the target response.

3.3.5. Out-of-distribution performance for inverse design

Following the in-distribution analysis, an extrapolation case-study of the data-driven design framework's performance on the validation set is now presented, by using responses generated from purely periodic structures. The evaluation follows the same best, median, and worst-case format. Periodic structures present a challenge, as their responses can differ greatly from the disordered metamaterials in the training set.

To quantify the degree of extrapolation, we calculate the NRMSE between each target response and its nearest neighbor within the training set. This metric represents the distance of a design target from the known training distribution. While the validation set for in-distribution tasks shows an average distance of 4.23%, the targets derived from periodic structures exhibit larger deviations, with distances ranging up to 40.04%. This quantitative gap confirms that these targets lie beyond the training bound.

Starting with the best-case result, an upper-bound extrapolation test, is shown in Fig. 14a. The target response for this case exhibits a nearest-neighbor distance of 31.55%, indicating a substantial shift from the disordered training data. Despite this distributional gap, the optimal

design perfectly reproduces the periodic reference structure. This successful outcome demonstrates the model's powerful ability to extrapolate toward high-performance targets.

The median-case is shown in Fig. 14b, which acts as an extrapolation test in the structural domain but an interpolation test in the performance domain. Quantitatively, the target response exhibits a nearest-neighbor distance of 5.31% relative to the training set, which is comparable to the in-distribution baseline of 4.23%. This metric confirms that the performance requirement remains within the effective interpolation range of the model, explaining the accurate prediction. The optimal design produces a disordered structure to achieve a similar response.

Conversely, the worst-case result represents a lower-bound extrapolation test, as shown in Fig. 14c. Both the optimal and suboptimal designs exhibit errors, erroneously incorporating high-stiffness cells instead of consistently utilizing low-stiffness units. Quantitatively, the target yields a nearest-neighbor distance of 30.00%, confirming that it lies far beyond the training distribution. However, unlike the successful upper-bound extrapolation which had a similar distance, this case fails due to the specific nature of the target. The response approaches a near-

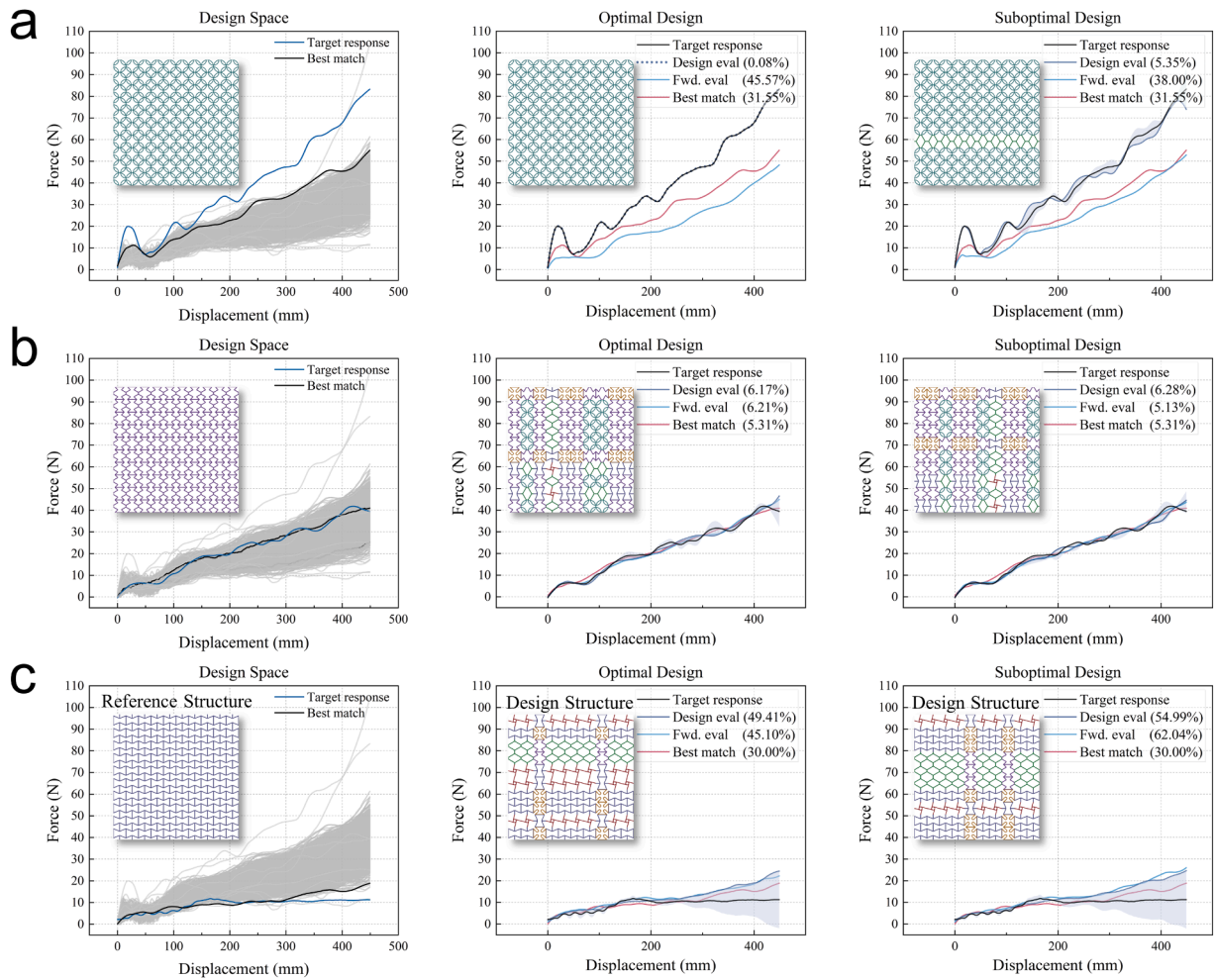


Fig. 14. Case-study of inverse design extrapolation performance on targets derived from periodic structures. For each case, the first column shows the periodic reference structure and its target response curve relative to the design space, while the second and third columns present the optimal and suboptimal designs generated by the model. The rows display the results for the (a) best-case, (b) median-case, and (c) worst-case scenarios to analyze the model's extrapolation capabilities. Legend percentages represent NRMSE values, while the shaded area shows the deviation between the evaluated design response and the target response.

zero stiffness condition, a physical extreme that is virtually absent from the training dataset. Consequently, the model lacks the learned patterns to approximate such negligible load-bearing behaviors and fails to extrapolate toward this lower performance boundary. Future work must address this bias by introducing training samples that cover these extreme mechanical properties.

4. Concluding remarks

This study presents a data-driven framework for the inverse design of disordered metamaterials with tailored nonlinear responses. At its core, a generative AR model is seamlessly integrated with a FNO as a fast surrogate model. The AR model's causal, sequential generation process inherently overcomes the critical challenge of structural connectivity. Our framework resolves the dual problem of guaranteeing connectivity while achieving target nonlinear performance. The main contributions of this work are threefold:

1. An autoregressive inverse design ensuring structural connectivity. We introduce an AR model that overcomes the connectivity challenge by learning implicit rules from data. It sequentially generates structurally valid disordered metamaterials while approaching the target response.
2. Quantifying the trade-off between performance and connectivity. Our work reveals the intrinsic relationship between achieving target

performance and maintaining structural connectivity, providing designers with direct control over this balance during the inverse design process.

3. An efficient data-driven framework for on-demand design. Our framework integrates the AR inverse model and the FNO surrogate into a closed loop, predicting the performance for disordered metamaterials that meet target nonlinear performance, thereby achieving high design efficiency.

Moving forward, this work opens compelling research avenues. This autoregressive framework establishes a foundation for several research trajectories. Integrating the sequential logic of transformer architectures with the generative capacity of diffusion models represents an immediate extension to enhance morphological diversity while ensuring structural connectivity. This paradigm further enables multi-modal co-design where structural geometry and material properties are optimized simultaneously through unified sequence modeling. These integrated methodologies facilitate the autonomous design of complex disordered systems with target performance for advanced engineering applications.

CRedit authorship contribution statement

Dazhi Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yujie Xiang:** Writing –

review & editing, Visualization, Software, Methodology, Conceptualization. **Peng Zhang:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Ning Liu:** Writing – review & editing, Supervision. **Xianqiao Wang:** Writing – review & editing, Visualization, Supervision. **Keke Tang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work was funded by the National Natural Science Foundation of China (No. 11872278), the Natural Science Foundation of Shanghai (Grant No. 25ZR1401355), and the Fundamental Research Funds for the Central Universities.

Appendix

1. Analysis of attention maps

To gain deeper insight into the model's decision-making process, heatmaps are visualized from different attention heads during the generation of the same design output described above. During the generation of each unit cell, the model refers to the target nonlinear response curve and the structural history. Sequential generation in this framework is not equivalent to local reasoning. Every local decision is made under the constraints of the global context. As a Causal Multi-head Attention mechanism is employed, the resulting heatmaps have a lower triangular shape. The vertical axis represents the Query (Q) from the current token being generated, while the horizontal axis represents the Key (K) from all previous tokens, including the initial condition. A high attention score indicates that the model heavily weighs the information from a previous token when making its current decision.

The analysis reveals five distinct attention patterns, which represent different reasoning strategies learned by the model. The first pattern is a long-range strategy shown in Fig. A1a. Here, each token attends to previous tokens within the same vertical column. This suggests the model is focusing on rules related to vertical connectivity between unit cells. The second pattern is a global strategy shown in Fig. A1b. The first column of the heatmap is intensely activated, showing that nearly every token generation step strongly attends to the initial performance target. This shows that the entire design process is guided by the conditional input. The third pattern illustrates a local strategy shown in Fig. A1c. High attention scores are concentrated along the main diagonal. This indicates that each decision is primarily influenced by the few tokens generated immediately before it, likely to ensure horizontal connectivity within a row. The fourth pattern is also a long-range strategy shown in Fig. A1d. At the beginning of each new row, which occurs every ten tokens, the model focuses intensely on the tokens of the very first row. This shows the model has learned that the layout of the first row critically constrains the vertical connectivity for the entire structure. The fifth pattern is a hybrid strategy shown in Fig. A1e. This attention head combines local, global and long-range patterns.

To establish a rigorous baseline for the quantitative attention analysis, we define a uniform causal prior over the structure-to-structure history. Let N denote the number of structural tokens and let $i \in \{0, \dots, N-1\}$ denote the query index in raster order. The valid keys for query i are $\{0, \dots, i\}$, so the uniform baseline weight is:

$$w_{i \rightarrow j} = \frac{1}{i+1}, \quad j \leq i. \quad (19)$$

We report the global baseline as the mean over all queries:

$$\bar{w} = \frac{1}{N} \sum_{i=0}^{N-1} \frac{1}{i+1} \quad (20)$$

For spatial-neighbor baselines on a grid of width W , the left-neighbor queries satisfy $i \bmod W \neq 0$ and the top-neighbor queries satisfy $i \geq W$. The baselines are:

$$\bar{w}_{\text{left}} = \frac{1}{|S_{\text{left}}|} \sum_{i \in S_{\text{left}}} \frac{1}{i+1}, \quad \bar{w}_{\text{up}} = \frac{1}{|S_{\text{up}}|} \sum_{i \in S_{\text{up}}} \frac{1}{i+1} \quad (21)$$

where $S_{\text{left}} = \{i | i \bmod W \neq 0\}$ and $S_{\text{up}} = \{i | i \geq W\}$.

To provide a quantitative basis for these visual patterns, we analyze the attention distribution to explicitly link the model's focus with mechanical targets and physical connectivity constraints. We extract the raw attention matrices from the inference phase and decouple the contributions of the conditional prompt from the structure-to-structure causal history through selective masking and renormalization. regarding mechanical characteristics, we compute the average attention weight assigned to the conditional prompt token across all structural tokens. While the global mean score across all layers and heads is 0.074, the distribution exhibits specialization. Specifically, head 5 in layer 0 yields a score of 0.89, representing a 12.0-fold increase over the global average. Regarding physical connectivity, we isolate structure-to-structure attention by masking the conditional and start tokens to re-normalize the weights. We then compare the attention allocated to spatial neighbors against a baseline uniform distribution over the causal history. Under the uniform assumption, the expected probabilities for the immediate left and top neighbors are 4.35% and 2.51%, respectively. In contrast, the model exhibits average observed probabilities of 6.09% and 3.22% for these positions, corresponding to elevation factors of 1.40 and 1.28 relative to the baseline.

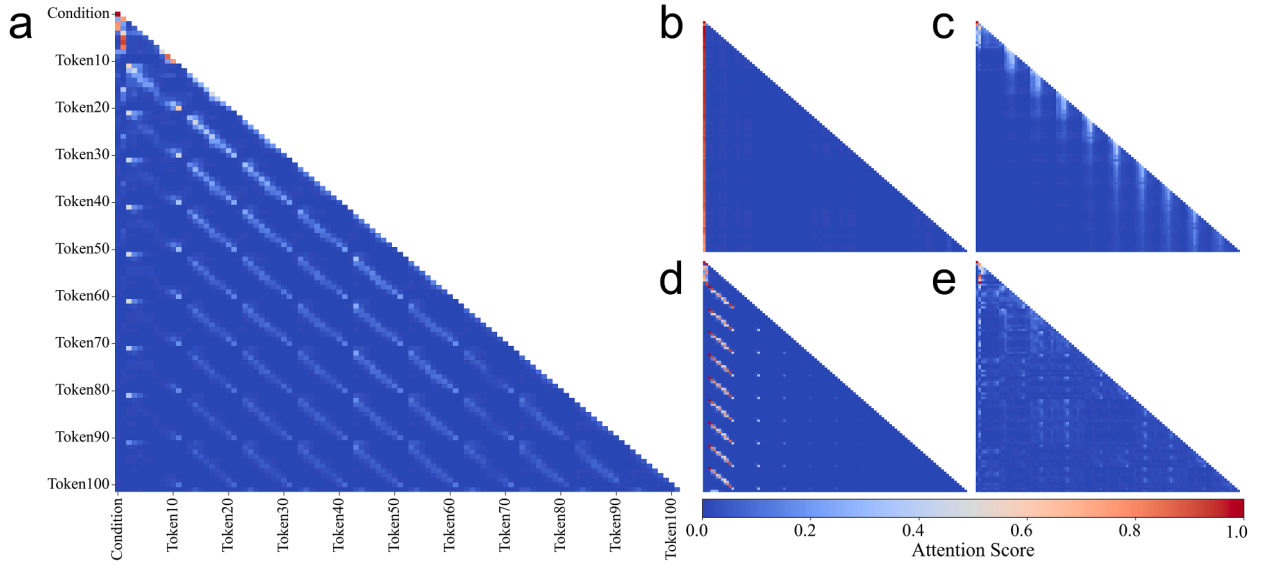


Fig. A1. Visualization of the attention maps learned by the AR model. The attention maps reveal the model's reasoning by showing the influence of previous tokens on the generation of the current one. We identify several distinct strategies: a local strategy (c), which prioritizes adjacent unit cells to maintain connectivity; a global strategy (b), which constantly references the target performance condition; and long-range strategies (a, d), which capture initial-sequence dependencies. And the model also employs a hybrid strategy (e), integrating local, global and long-range information.

2. Performance evaluation across different generation temperatures

Fig. A2 displays the design results, including the performance predicted by the FNO and the best matching curve from the training set. For any design, the primary goal is to minimize the NRMSE between its actual simulated performance, the Design eval, and the Target response. The FNO's prediction, or Fwd. eval, serves as a rapid performance estimate within the design loop. The accuracy of the FNO itself is judged by the closeness between the Fwd. eval and Design eval curves. A large deviation of both curves from the target indicates a limitation of the inverse design at that temperature, not necessarily a failure of the forward model.

The influence of the temperature parameter t is now analyzed. At the lowest viable temperature of $t = 0.25$, the model's choices are conservative, resulting in a design output that is highly repetitive and approaches a periodic structure. This occurs because low temperatures limit exploration, causing the model to repeatedly select the safest, most probable unit cells. As the temperature increases to $t = 0.5$, the design output exhibits greater disorder and structural diversity. This leads to the globally best performance, as the design accurately captures the target's peaks and valleys with minimal error. Its performance even surpasses that of the best match from the training set. We define this result as the optimal design. A further increase to $t = 0.75$ results in a performance deviation. The model incorrectly places Chiral unit cells in a critical load-bearing position. While Chiral units can be effective, they are also prone to in-plane buckling, which causes a sharp drop in stiffness and maximum force. This failure produces the highest NRMSE among all temperatures, and we therefore define it as the worst design. When the temperature is set to $t = 1$, the performance improves again. The resulting design output has the second-lowest NRMSE, defined as the suboptimal design. The final two cases at $t = 1.25$ and $t = 1.5$ show considerable performance errors but do not exhibit the catastrophic stiffness collapse seen in the worst design. Notably, all six designs presented in this study maintain perfect connectivity, confirming the model's ability to adhere to the connectivity constraint while exploring the design space.

To quantify the topological deviation of the generated designs from the reference structure, a structural similarity metric $\mathcal{S}$ is introduced. This metric is defined as the fraction of grid positions where the generated unit cell type matches the reference unit cell type exactly. For a 10×10 grid, the structural distance $\mathcal{D}$ and similarity $\mathcal{S}$ are calculated as:

$$\mathcal{D} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(c_{\text{gen}}^{(i)} \neq c_{\text{ref}}^{(i)}), \quad \mathcal{S} = 1 - \mathcal{D} \quad (22)$$

where $N = 100$ is the total number of unit cells, $c_{\text{gen}}^{(i)}$ and $c_{\text{ref}}^{(i)}$ denote the unit cell indices at position i for the generated and reference structures, respectively, and $\mathbb{I}$ is the indicator function.

Quantitative analysis indicates that structural similarity remains low, ranging from 0.03 to 0.16, across the temperature spectrum. Notably, the design with optimal mechanical performance exhibits a similarity of only 0.03, representing a 97% topological difference from the reference. This result demonstrates that the model prioritizes the accuracy of the mechanical response over geometric replication. The framework successfully matches the target nonlinear curve with low NRMSE by exploiting mechanically equivalent solutions within the design space. For instance, distinct topologies like Re-entrant and Bowtie structures possess analogous stiffness characteristics. The model learns to substitute these equivalent units to construct alternative architectures that satisfy global performance requirements, thereby navigating the non-uniqueness of the structure-property mapping rather than memorizing specific layouts.

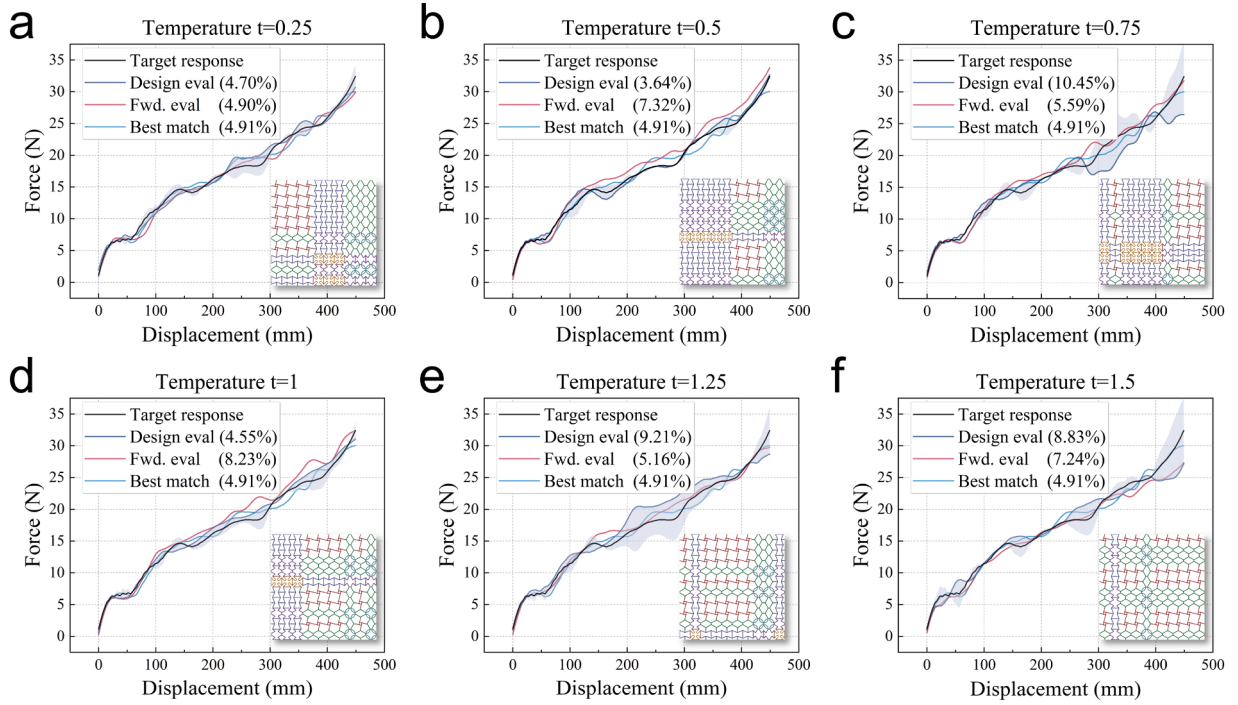


Fig. A2. Performance evaluation of inverse designs generated at various temperatures for a single target response. The figure shows six design results, (a) through (f), generated by the AR model using different temperature settings for the same target curve from Fig. 10. Each panel displays the generated structure (inset) and a plot comparing the desired goal (Target response) with three other key curves: the actual simulated performance of the generated structure (Design eval), the performance predicted by the forward prediction model (Fwd. eval), and the closest response available in the training set (Best match). Legend percentages represent NRMSE values, while the shaded area shows the deviation between the evaluated design response and the target response.

3. Baseline model comparison and training analysis

To assess the training stability and rule out overfitting given the dataset size, we conduct a comprehensive evaluation comparing the proposed autoregressive Transformer against standard baseline architectures, including cVAE, GAN, MLP, and RNN. The specific configurations for these models, including their training objectives and output modalities, are detailed in Table A1. We monitor the learning dynamics of each model by tracking the loss history on both the training and validation sets.

Table A1

Summary of the model architectures used for baseline comparison.

Model	Autoregressive	Task	Loss	Output
cVAE	×	Reg	MSE + KL	Continuous
GAN	×	Reg	MSE	Continuous
MLP	×	Cls	CE	Discrete
RNN	✓	Cls	CE	Discrete
Transformer	✓	Cls	CE	Discrete

As illustrated in Fig. A3, the learning curves demonstrate distinct convergence behaviors across the different methods. The Transformer model exhibits a rapid decrease in loss during the initial epochs, followed by a stable convergence phase. Crucially, the validation loss maintains a tight alignment with the training loss throughout the entire process, with no divergence observed even after extended training epochs. This minimal generalization gap confirms that the Transformer effectively captures the underlying structural rules without memorizing the training data, thereby addressing concerns regarding potential overfitting on the dataset. In contrast, baseline models such as the MLP show a slower convergence rate and a more noticeable disparity between training and validation performance, further highlighting the data efficiency and robustness of the proposed attention-based framework for this specific inverse design task.

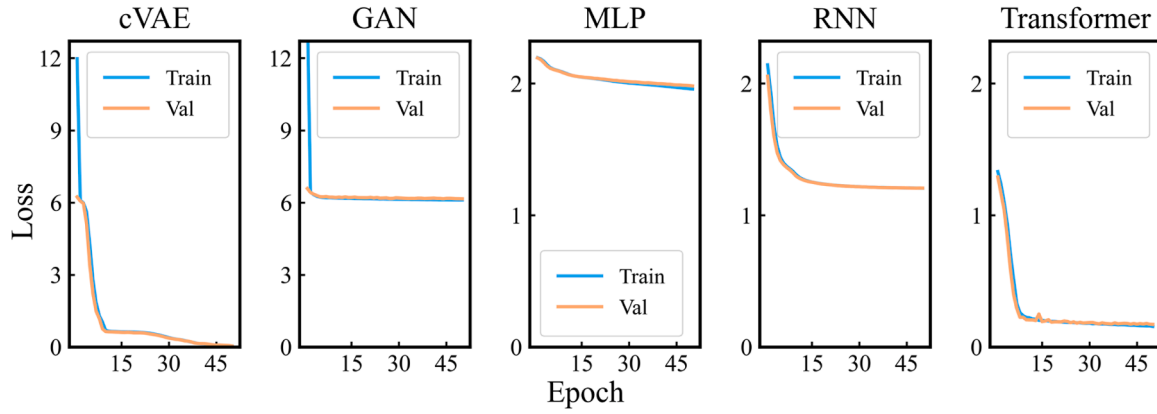


Fig. A3. Training and validation loss curves for the Transformer model and baseline models. The plots display the evolution of loss values over training epochs, illustrating the convergence behavior and generalization gap for each architecture.

4. Simulation, training, and inference time and device reports

All computational tasks are performed on a workstation equipped with an Intel(R) Xeon(R) Platinum 8350C CPU and an NVIDIA GeForce RTX 4090 GPU. [Table A2](#) summarizes the time consumption for numerical simulation, model training, and inference. The generation of a single data sample via FEA requires approximately 1010.8 s on average due to the complex contact and nonlinear calculations involved. In contrast, the trained FNO surrogate model predicts the nonlinear response curve in approximately 0.006 s per sample. This represents a speedup of roughly five orders of magnitude compared to the numerical simulation, reducing the cost of forward performance evaluation. For the inverse design task, the AR model generates a valid structural design in approximately 1.0 s. A traditional optimization approach typically requires hundreds or thousands of iterative FEA calls to converge to a single solution, potentially taking days. The proposed framework, once trained, bypasses these iterations and directly outputs the design, demonstrating improvement in computational efficiency for on-demand design tasks.

Table A2

Computational resources and time costs for data generation, model training, and inference.

Category	Activity	Time
Simulation	FEA simulation	~ 1010.8 s/ sample
Training	AR training	~ 301.2 s
	FNO training	~ 2752.4 s
Inference	AR inference	~ 1.0 s/ sample
	FNO inference	~ 0.006 s/ sample

5. Top-P sampling details and parameter analysis

The sensitivity analysis presented in [Fig. A4](#) quantifies how the sampling hyperparameters influence the stability of the generative process. Reducing the Top-P threshold or the temperature restricts the candidate pool to a narrow subset of unit cells with the highest predicted probabilities. While this conservative sampling strategy ensures high structural connectivity by strictly adhering to the most likely local patterns, it inadvertently limits the morphological diversity of the generated designs. This excessive restriction prevents the model from exploring valid but less probable structural configurations that are essential for matching complex nonlinear responses. Conversely, increasing the Top-P threshold toward 1.0 incorporates lower-probability tokens into the sampling pool, which introduces stochastic noise and degrades the mean connectivity rate, particularly at higher temperatures. Consequently, a Top-P value of 0.95 was selected for this study. This threshold effectively filters out the implausible tail of the probability distribution while retaining sufficient stochasticity to maintain an equilibrium between exploiting learned structural rules for validity and exploring the design space for diversity.

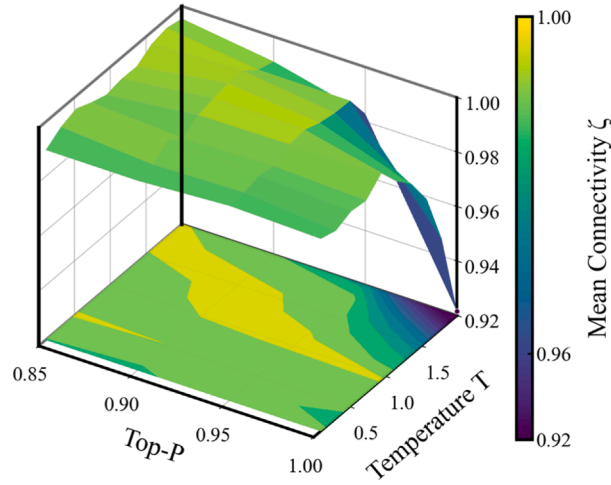


Fig. A4. The sampling parameters surface plot. It displays the mean connectivity rate as a function of the Top-P probability threshold and the temperature parameter.

6. Quantitative analysis of mesh convergence

To verify the numerical reliability, we perform a quantitative mesh sensitivity analysis, shown in [Table A3](#). The model with the finest mesh size of 1.75 mm serves as the ground truth benchmark. This configuration consists of 61,543 elements and represents the highest fidelity level in our study. We assess convergence based on two scalar metrics: the maximum reaction force and the total energy absorption. The results indicate that the 2.0 mm mesh size achieves a high degree of accuracy relative to the benchmark. Specifically, the relative error for the maximum reaction force is 0.6%. The deviation for the energy absorption is also limited to a maximum of 1.6% across the tested range. However, increasing the mesh size to 2.75 mm leads to a noticeable divergence, where the force error rises to 6.4%. Therefore, we adopt the 2.0 mm mesh for all simulations, as it effectively balances computational speed with sufficient numerical precision.

Table A3

Mesh convergence analysis results.

Mesh size	Element count	Max force (N)	Energy (J)	Error force (%)	Error energy (%)
1.75	61,543	43.7	8862.7	0.0	0.0
2.00	52,287	44.0	8950.9	0.6	1.0
2.25	37,208	44.7	9003.2	2.1	1.6
2.75	26,029	46.5	8956.5	6.4	1.1

7. Analysis of design space coverage ratio

To quantify the proportion of the design space covered by the dataset, we introduce a coverage ratio metric, η . Let D represent the theoretical design space bounded by the limit mechanical responses of the unit cell library, and R represent the envelope of the 2423 training samples. The coverage ratio is defined as the area of the intersection between the training envelope and the theoretical space divided by the total theoretical area:

$$\eta = \frac{\text{Area}(D \cap R)}{\text{Area}(D)} \quad (23)$$

The coverage situation is shown in [Fig. A5](#). The calculated coverage ratio for the current dataset is approximately 60.4%. This indicates that the training data occupies the majority of the effective mechanical property space, while the remaining region represents the potential design space for the generative model to explore. The sufficiency of this dataset scale is validated by the extrapolation performance discussed in [Section 3.3.5](#). The model successfully generates designs for periodic structures that lie at the boundaries of the design space and are absent from the training set. This capability to generalize to out-of-distribution targets confirms that the dataset size is sufficient for the model to learn the underlying mapping relationship between disordered structures and complex nonlinear mechanical responses.

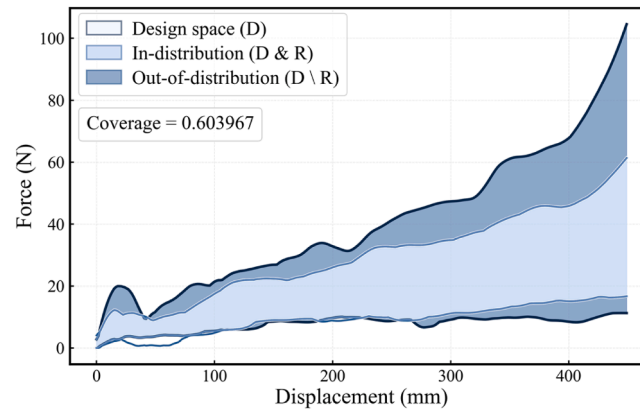


Fig. A5. Visualization of the design space coverage by the training dataset. The region D defines the theoretical design space achievable by the unit cell library, while R represents the response envelope of the training set. The intersection $D \cap R$ indicates the in-distribution region covered by training samples, and $D \setminus R$ denotes the out-of-distribution region. The design space coverage ratio is calculated as the area of $D \cap R$ divided by the total area of D .

Data Availability

All data and code required to reproduce the findings of this study have been deposited in the GitHub repository at: <https://github.com/dazhizhao/Autoregressive-Inverse-Design-for-Disordered-Metamaterials>.

References

- [1] P. Jiao, J. Mueller, J.R. Raney, X. Zheng, A.H. Alavi, Mechanical metamaterials and beyond, *Nat. Commun.* 14 (2023) 6004.
- [2] K. Bertoldi, V. Vitelli, J. Christensen, M. Van Hecke, Flexible mechanical metamaterials, *Nat. Rev. Mater.* 2 (2017) 1–11.
- [3] B. Florijn, C. Coulais, M. van Hecke, Programmable mechanical metamaterials, *Phys. Rev. Lett.* 113 (2014) 175503.
- [4] T.A. Schaedler, W.B. Carter, Architected cellular materials, *Annu. Rev. Mater. Res.* 46 (2016) 187–210.
- [5] W. Lin, Y. Yan, S. Zhao, H. Qin, Y. Liu, Digital mechanical metamaterial with programmable functionality, *Adv. Mater.* 36 (2024) 2406263.
- [6] Y. Wei, F. Pan, X. Lin, L. Zhang, J. Xiang, Y. Chen, On-demand reprogrammable mechanical metamaterial driven by structure performance relations, *Adv. Mater.* 37 (2025) 2410865.
- [7] W. Liu, S. Janbaz, D. Dykstra, B. Ennis, C. Coulais, Harnessing plasticity in sequential metamaterials for ideal shock absorption, *Nature* 634 (2024) 842–847.
- [8] Y. Wang, Q. Zeng, J. Wang, Y. Li, D. Fang, Inverse design of shell-based mechanical metamaterial with customized loading curves based on machine learning and genetic algorithm, *Comput. Methods Appl. Mech. Eng.* 401 (2022) 115571.
- [9] B. Deng, A. Zareei, X. Ding, J.C. Weaver, C.H. Rycroft, K. Bertoldi, Inverse design of mechanical metamaterials with target nonlinear response via a neural accelerated evolution strategy, *Adv. Mater.* 34 (2022) 2206238.
- [10] C.S. Ha, D. Yao, Z. Xu, C. Liu, H. Liu, D. Elkins, M. Kile, V. Deshpande, Z. Kong, M. Bauchy, Rapid inverse design of metamaterials based on prescribed mechanical behavior through machine learning, *Nat. Commun.* 14 (2023) 5765.
- [11] C. Fox, K. Chen, M. Antonini, T. Magrini, C. Daraio, Extracting geometry and topology of orange pericarps for the design of bioinspired energy absorbing materials, *Adv. Mater.* 36 (2024) 2405567.
- [12] Y. Lei, Y. Wang, R. Cui, X. Huang, L. Zhang, Y. Jin, J. Gao, B. Deng, 3D Hyperbolic kirigami metamaterials with tunable auxeticity and multistability, *Adv. Sci.* (2025) e06703.
- [13] Y. Xiang, J. Hou, X. Chen, R. Pidaparti, K. Song, K. Tang, X. Wang, A GAN-based stepwise full-field mechanical prediction model for architected metamaterials, *Int. J. Mech. Sci.* 284 (2024) 109771.
- [14] K. Tang, J. Tian, Y. Xiang, Z. Zhong, X. Wang, Adaptive triangular-mesh coarse-grained model for notched 2D metamaterials: a hybrid FEA and top-down approach, *Theoret. Appl. Fract. Mech.* 127 (2023) 104077.
- [15] Y. Xiang, J. Tian, K. Tang, X. Wang, Z. Zhong, Morphological design and tunable mechanical properties of 3D spinodal membrane structures: adaptive coarse-grained modelling, *Acta Mech. Sinica* 40 (2024) 423655.
- [16] S. Xiang, Z. Dong, B. Jia, B. Chen, H. Huang, A new hybrid TPMS design with excellent thermo-mechanical performance, *Int. J. Mech. Sci.* (2025) 111132.
- [17] J. Yan, Y. Gao, S. Xiang, J. Zhan, L. Jin, H. Huang, A strategy for 3D printed Al-based Bi-material lattice optimization: thermal-mechanical coupling performance with load-bearing and energy-absorption, *Adv. Funct. Mater.* (2025) e21424.
- [18] M. Zaiser, S. Zapperi, Disordered mechanical metamaterials, *Nat. Rev. Phys.* 5 (2023) 679–688.
- [19] S. Choukir, N. Manohara, C.V. Singh, Disorder unlocks the strength-toughness trade-off in metamaterials, *Appl. Mater. Today* 42 (2025) 102579.
- [20] J. Huang, S. Xu, Y. Ma, J. Liu, A topology optimization method for hyperelastic porous structures subject to large deformation, *Int. J. Mech. Mater. Design* 18 (2022) 289–308.
- [21] T. Xue, S. Mao, Mapped shape optimization method for the rational design of cellular mechanical metamaterials under large deformation, *Int. J. Numer. Methods Eng.* 123 (2022) 2357–2380.
- [22] L. Wang, Co-design of magnetic soft robots with large deformation and contacts via material point method and topology optimization, *Comput. Methods Appl. Mech. Eng.* 445 (2025) 118205.
- [23] L. Wang, A.L. Evenchik, J.M. Yang, R.L. Truby, W. Chen, Autonomous codesign and fabrication of multistimuli-responsive material systems, *Sci. Adv.* 11 (2025) ead4409.
- [24] N. Elango, A.A.M. Faudzi, A review article: investigations on soft materials for soft robot manipulations, *Int. J. Adv. Manuf. Technol.* 80 (2015) 1027–1037.
- [25] J. Zhang, G. Lu, Z. You, Large deformation and energy absorption of additively manufactured auxetic materials and structures: a review, *Compos. Part B: Eng.* 201 (2020) 108340.
- [26] Y.-C. Chan, D. Da, L. Wang, W. Chen, Remixing functionally graded structures: data-driven topology optimization with multiclass shape blending, *Struct. Multidisc. Optim.* 65 (2022) 135.
- [27] D. Lee, W. Chen, L. Wang, Y.C. Chan, W. Chen, Data-driven design for metamaterials and multiscale systems: a review, *Adv. Mater.* 36 (2024) 2305254.
- [28] Z. Yang, J. Yi, F. Li, Z. Li, L. Ye, B. Li, J. Christensen, Static mechanical cloaking and camouflage from disorder, *Nat. Commun.* 16 (2025) 8858.
- [29] K. Liu, R. Sun, C. Daraio, Growth rules for irregular architected materials with programmable properties, *Science* 377 (2022) 975–981.
- [30] Y. Jia, K. Liu, X.S. Zhang, Unstructured growth of irregular architectures for optimized metastructures, *J. Mech. Phys. Solids* 192 (2024) 105787.
- [31] L. Wang, Y.-C. Chan, F. Ahmed, Z. Liu, P. Zhu, W. Chen, Deep generative modeling for mechanistic-based learning and design of metamaterial systems, *Comput. Methods Appl. Mech. Eng.* 372 (2020) 113377.
- [32] Z. Chai, Z. Zong, H. Yong, X. Ke, J. Zhu, H. Ding, C.F. Guo, Z. Wu, Tailoring stress-strain curves of flexible snapping mechanical metamaterial for on-demand mechanical responses via data-driven inverse design, *Adv. Mater.* 36 (2024) 2404369.
- [33] P. Zhang, K. Tang, A. Wang, H. Wu, Z. Zhong, Neural network integrated with symbolic regression for multiaxial fatigue life prediction, *Int. J. Fatigue* 188 (2024) 108535.
- [34] K. Tang, P. Zhang, Y. Zhao, Z. Zhong, Deep learning-based semantic segmentation for morphological fractography, *Eng. Fract. Mech.* 303 (2024) 110149.
- [35] P. Zhang, K. Tang, G. Chen, J. Li, Y. Li, Multimodal data fusion enhanced deep learning prediction of crack path segmentation in CFRP composites, *Compos. Sci. Technol.* 257 (2024) 110812.
- [36] P. Zhang, K. Tang, A TCN-based feature fusion framework for multiaxial fatigue life prediction: bridging loading dynamics and material characteristics, *Int. J. Fatigue* 197 (2025) 108915.
- [37] P. Zhang, A. Wang, K. Tang, Cross-material transfer learning for multiaxial fatigue prediction: a data-efficient adaptive framework, *Eng. Fract. Mech.* (2025) 111372.
- [38] X. Zhou, P. Zhang, Y. Jin, N. Liu, X. Wang, K. Tang, Classification-guided two-stage segmentation of multi-feature MA islands in bainitic steels, *Mater. Charact.* (2025) 115576.
- [39] P. Zhang, X. Zhou, R. Liang, J. Li, K. Tang, Y. Li, Multi-scale semantic segmentation for fiber identification and 3D reconstruction of unidirectional composite, *Compos. Sci. Technol.* 265 (2025) 111160.
- [40] B. Wang, P. Zhang, Y. Xiang, D. Wang, B. Wu, X. Wang, K. Tang, A. Chen, Advancing structural failure analysis with physics-informed machine learning in engineering applications, *Engineering* (2025).

- [41] P. Zhang, Y. Xiang, K. Tang, Deep learning surrogate for phase field modeling in fiber-reinforced composites: from stress evolution to final crack path, *Comput. Methods Appl. Mech. Eng.* 448 (2026) 118518.
- [42] Y. Zhao, Y. Xiang, K. Tang, Machine learning-based fatigue life prediction of lamellar titanium alloys: a microstructural perspective, *Eng. Fract. Mech.* 303 (2024) 110106.
- [43] W. Lu, N.A. Lee, M.J. Buehler, Modeling and design of heterogeneous hierarchical bioinspired spider web structures using deep learning and additive manufacturing, *Proc. Natl Acad. Sci.* 120 (2023) e2305273120.
- [44] J. Tian, M.T. Sobczak, D. Patil, J. Hou, L. Pang, A. Ramanathan, L. Yang, X. Chen, Y. Golan, X. Zhai, CrossMatAgent: AI-assisted design of manufacturable metamaterial patterns via multi-agent generative framework, *Adv. Intell. Discov.* (2025) 202500063.
- [45] J. Hou, X. Chen, T. Wu, E. Kuhl, X. Wang, Automated data-driven discovery of material models based on symbolic regression: a case study on the human brain cortex, *Acta Biomater.* 188 (2024) 276–296.
- [46] J. Hou, N. Filla, X. Chen, M.J. Razavi, D. Zhu, T. Liu, X. Wang, Exploring hyperelastic material model discovery for human brain cortex: multivariate analysis vs. artificial neural network approaches, *J. Mech. Behav. Biomed. Mater.* 165 (2025) 106934.
- [47] L. Jin, S. Yu, J. Cheng, H. Ye, X. Zhai, J. Jiang, K. Zhang, B. Jian, M. Bodaghi, Q. Ge, Machine learning driven forward prediction and inverse design for 4D printed hierarchical architecture with arbitrary shapes, *Appl. Mater. Today* 40 (2024) 102373.
- [48] L. Jin, S. Yu, J. Cheng, Z. Liu, K. Zhang, S. Zhou, X. He, G. Xie, M. Bodaghi, Q. Ge, Machine learning powered inverse design for strain fields of hierarchical architectures, *Compos. Part B: Eng.* 299 (2025) 112372.
- [49] S. Zhou, K. Zhang, L. Jin, Q. Gao, W.-H. Liao, Efficient data driven optimization framework for designing B-spline honeycombs with excellent energy absorption, *Thin-Walled Struct.* 209 (2025) 112941.
- [50] X.-c. Zhang, Z.-y. Song, Y.-n. Li, L.-j. Xiao, Z. Xu, L.-x. Rao, T.-j. Ci, X.-L. Hui, Generative inverse design of metamaterials with customized stress-strain response, *Int. J. Mech. Sci.* (2025) 110875.
- [51] L. Xiao, G. Shi, W. Song, Machine learning predictions on the compressive stress-strain response of lattice-based metamaterials, *Int. J. Solids Struct.* 300 (2024) 112893.
- [52] L. Jin, K. Zhang, S. Zhou, G. Xie, W.-H. Liao, Modulus tunability in hierarchical architectures: a machine learning-enabled approach, in: *Multifunctional Materials and Structures*, SPIE, 2025, pp. 133–143.
- [53] K. Tang, Y. Xiang, J. Tian, J. Hou, X. Chen, X. Wang, Z. Zhong, Machine learning-based morphological and mechanical prediction of kirigami-inspired active composites, *Int. J. Mech. Sci.* 266 (2024) 108956.
- [54] T. Xue, A. Beatson, M. Chirramonte, G. Roeder, J.T. Ash, Y. Menguc, S. Adriaenssens, R.P. Adams, S. Mao, A data-driven computational scheme for the nonlinear mechanical properties of cellular mechanical metamaterials under large deformation, *Soft. Matter* 16 (2020) 7524–7534.
- [55] T. Xue, S. Adriaenssens, S. Mao, Learning the nonlinear dynamics of mechanical metamaterials with graph networks, *Int. J. Mech. Sci.* 238 (2023) 107835.
- [56] E. Li, Y. Wang, L. Jin, Z. Zong, E. Zhu, B. Wang, Q. Wang, Z. Yang, W.-Y. Yin, Z. Wei, Current-diffusion model for metasurface structure discoveries with spatial-frequency dynamics, *Nat. Mach. Intell.* (2025) 1–11.
- [57] Z. Wei, J. Chen, K. Wei, Generative deep learning for designing irregular metamaterials with programmable nonlinear mechanical responses, *Int. J. Mech. Sci.* 291–292 (2025) 110123.
- [58] Y. Xiang, J. Hou, X. Chen, K. Tang, X. Wang, Decoupled design of hybrid mechanical metamaterials via ensembled deep learning, *Int. J. Mech. Sci.* (2025) 110514.
- [59] Y. Zhou, Y. Zheng, Y. Zhang, H. Wu, C. Liu, Inverse design of growth-inspired irregular architected materials for programmable properties, *Extreme Mech. Lett.* 70 (2024) 102196.
- [60] L. Wang, J. Boddapati, K. Liu, P. Zhu, C. Daraio, W. Chen, Mechanical cloak via data-driven aperiodic metamaterial design, *Proc. Natl Acad. Sci.* 119 (2022) e2122185119.
- [61] Y. Jia, K. Liu, X.S. Zhang, Modulate stress distribution with bio-inspired irregular architected materials towards optimal tissue support, *Nat. Commun.* 15 (2024) 4072.
- [62] Y. Jia, K. Liu, X.S. Zhang, Topology optimization of irregular multiscale structures with tunable responses using a virtual growth rule, *Comput. Methods Appl. Mech. Eng.* 425 (2024) 116864.
- [63] M. Shen, K. Liu, S. Mao, C. Daraio, Self-supervised AI for decoding and designing disordered metamaterials, *Sci. Adv.* 12 (2026) eadx7389.
- [64] J.-H. Bastek, D.M. Kochmann, Inverse design of nonlinear mechanical metamaterials via video denoising diffusion models, *Nat. Mach. Intell.* 5 (2023) 1466–1475.
- [65] Q. Liu, S. Koric, D. Abueidha, H. Meidani, P. Geubelle, Toward signed distance function based metamaterial design: neural operator transformer for forward prediction and diffusion model for inverse design, *Comput. Methods Appl. Mech. Eng.* 446 (2025) 118316.
- [66] Z. Zou, J. Liu, K. Gao, D. Chen, J. Yang, Z. Wu, Inverse design of functionally graded porous structures with target dynamic responses, *Int. J. Mech. Sci.* 280 (2024) 109530.
- [67] G. Felsch, V. Slesarenko, Generative models struggle with kirigami metamaterials, *Sci. Rep.* 14 (2024) 19397.
- [68] J. Xiong, G. Liu, L. Huang, C. Wu, T. Wu, Y. Mu, Y. Yao, H. Shen, Z. Wan, J. Huang, Autoregressive models in vision: a survey, *arXiv preprint arXiv:2411.05902* (2024).
- [69] P. Esser, R. Rombach, B. Ommer, Taming transformers for high-resolution image synthesis, in: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 12873–12883.
- [70] T. Li, Y. Tian, H. Li, M. Deng, K. He, Autoregressive image generation without vector quantization, *Adv. Neural Inf. Process. Syst.* 37 (2024) 56424–56445.
- [71] K. Tian, Y. Jiang, Z. Yuan, B. Peng, L. Wang, Visual autoregressive modeling: scalable image generation via next-scale prediction, *Adv. Neural Inf. Process. Syst.* 37 (2024) 84839–84865.
- [72] J. Mu, N. Vasconcelos, X. Wang, Editor: unified conditional generation with autoregressive models, in: *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 7899–7909.
- [73] K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M.R. Equi, C. Finn, N. Fusai, M.Y. Galliker, $\pi_{0.5}$: a vision-language-action model with open-world generalization, *9th Annual Conference on Robot Learning*, 2025.
- [74] M.J. Buehler, MeLM, a generative pretrained language modeling framework that solves forward and inverse mechanics problems, *J. Mech. Phys. Solids* 181 (2023) 105454.
- [75] R.K. Luu, M.J. Buehler, BioinspiredLLM: conversational large language model for the mechanics of biological and bio-inspired materials, *Adv. Sci.* 11 (2024) 2306724.
- [76] M.J. Buehler, Cephalo: multi-modal vision-language models for bio-inspired materials analysis and design, *Adv. Funct. Mater.* 34 (2024) 2409531.
- [77] K. Naghavi Khanghah, Z. Wang, H. Xu, Generating porous metamaterial designs using variational graph autoencoder and large language model, in: *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, American Society of Mechanical Engineers, 2024. V03AT03A007.
- [78] M. Maurizi, D. Xu, Y.-T. Wang, D. Yao, D. Hahn, M. Oudich, A. Satpati, M. Bauchy, W. Wang, Y. Sun, Designing metamaterials with programmable nonlinear responses and geometric constraints in graph space, *Nat. Mach. Intell.* (2025) 1–14.
- [79] J. Krmela, A. Artyukhov, V. Krmelová, O. Pozovnyi, Determination of material parameters of rubber and composites for computational modeling based on experiment data, *Journal of Physics: Conference Series*, IOP Publishing, 2021 012047.
- [80] E. Cao, Z. Dong, B. Jia, H. Huang, Inverse design of isotropic auxetic metamaterials via a data-driven strategy, *Mater. Horiz.* (2025).
- [81] N.B. Kovachki, S. Lanthaler, A.M. Stuart, Operator learning: algorithms and analysis, *Handbook Numer. Anal.* 25 (2024) 419–467.
- [82] H. Jin, B. Zhang, Q. Cao, E. Zhang, A. Bora, S. Krishnaswamy, G.E. Karniadakis, H. D. Espinosa, Characterization and inverse design of stochastic mechanical metamaterials using neural operators, *Adv. Mater.* (2025) 2420063.
- [83] D. Lee, L. Zhang, Y. Yu, W. Chen, Deep neural operator enabled concurrent multitask design for multifunctional metamaterials under heterogeneous fields, *Adv. Opt. Mater.* 12 (2024) 2303087.
- [84] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, *arXiv preprint arXiv:2010.08895* (2020).
- [85] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* (2017) 30.
- [86] L. McInnes, J. Healy, J. Melville, Umap: uniform manifold approximation and projection for dimension reduction, *arXiv preprint arXiv:1802.03426*, (2018).
- [87] M. Ahmed, R. Seraj, S.M.S. Islam, The k-means algorithm: a comprehensive survey and performance evaluation, *Electronics* 9 (2020) 1295.
- [88] O. Ronneberger, P. Fischer, T. Brox, U-net: convolutional networks for biomedical image segmentation. *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2015, pp. 234–241.